# Waterbouwkunde
## CTB2410

## Week 7

## Dikes and BreakWaters

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1 June, 2026



# Content

A. Introduction
B. What is a defence ?
C. How does it fail ?
D. Breakwaters
E. Bed Protection
F. Conclusions

# A

# Introduction

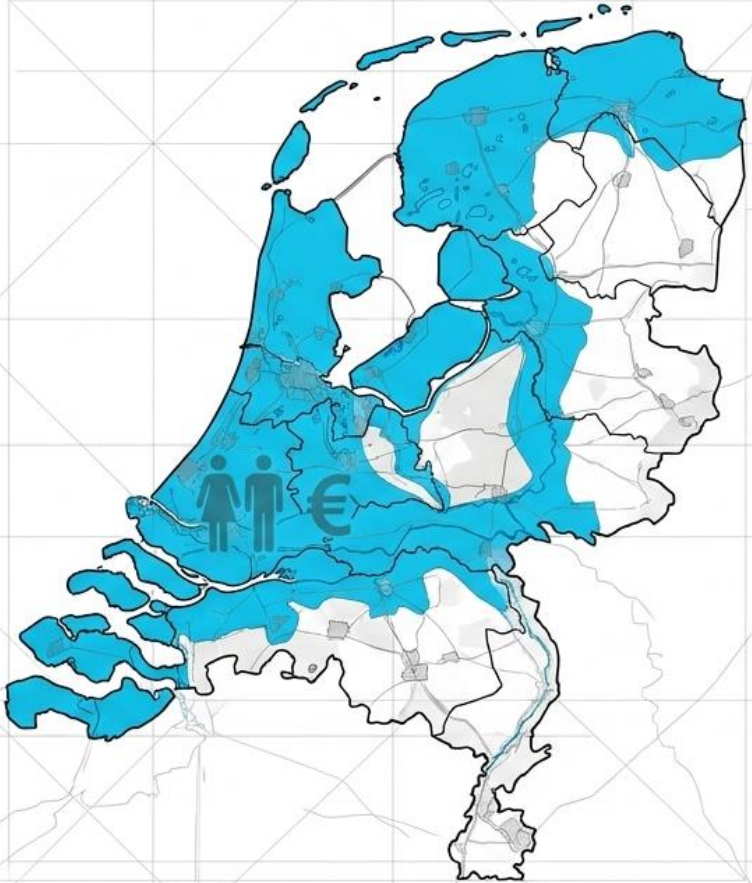
Structural Flood Resilience

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**Delft University of Technology**



***"We will be more flood resilient in the era of AI and sensors than ever"***

https://www.tudelft.nl/en/ceg/research/stories-of-science/making-dikes-safer-with-a-bass-guitar



**60%** of the Netherlands lies below mean sea level.

Without flood defenses, this entire **cyan area** would be permanently flooded.

**9 Million** people and **70%** of the Dutch GDP reside in this **vulnerable zone**.

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# A CENTURY OF ACTION AND REACTION

## 1916 ZUIDERZEE FLOOD

**Action:** Combined pluvial and fluvial storm breaks dozens of dikes; 19 casualties.

**Reaction:** Engineer Cornelis Lely's plan to construct the massive Afsluitdijk, physically closing off the Zuiderzee.

## 1926 RHINE & MEUSE FLOODING

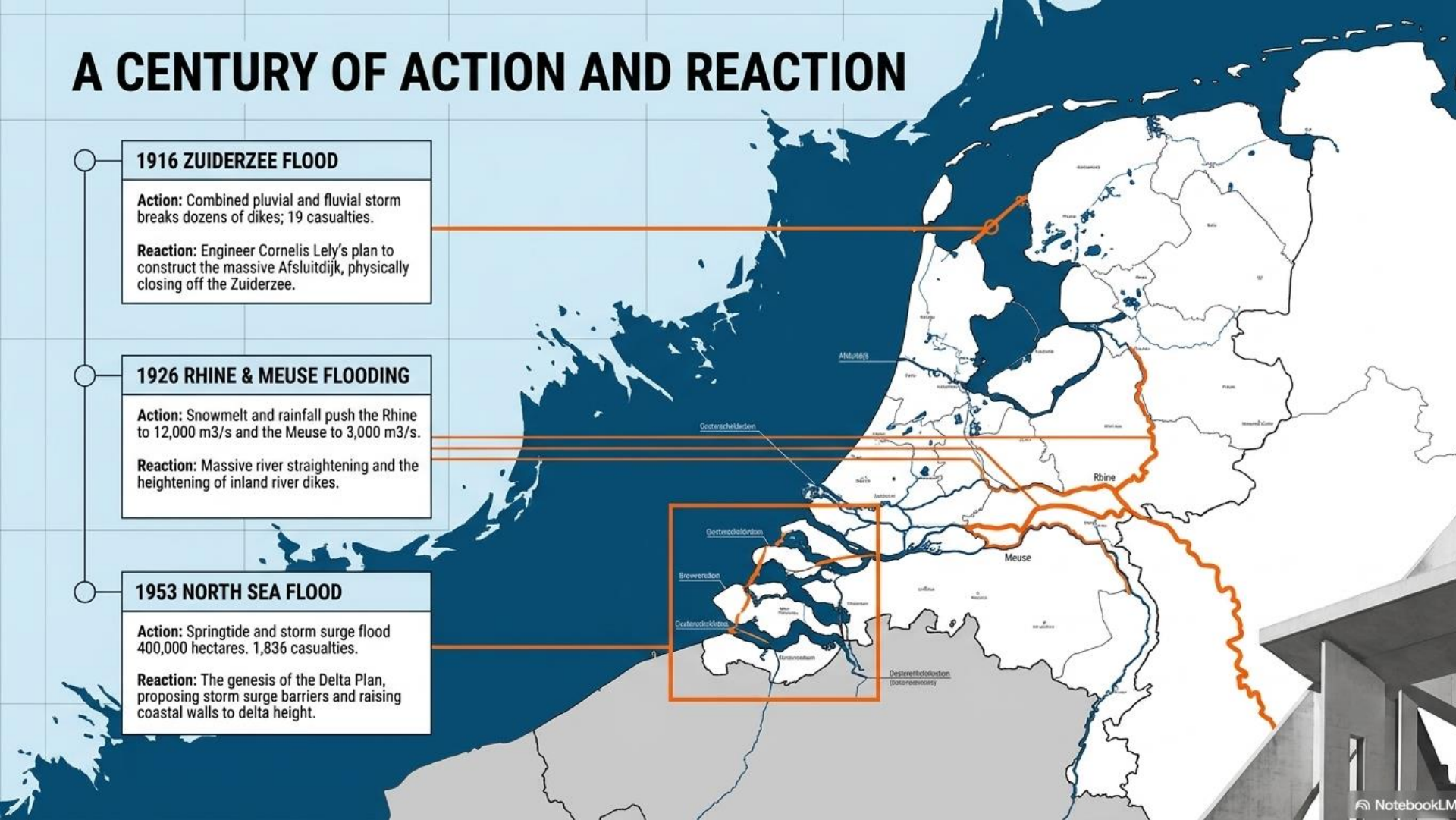
**Action:** Snowmelt and rainfall push the Rhine to 12,000 m3/s and the Meuse to 3,000 m3/s.

**Reaction:** Massive river straightening and the heightening of inland river dikes.

## 1953 NORTH SEA FLOOD

**Action:** Springtide and storm surge flood 400,000 hectares. 1,836 casualties.

**Reaction:** The genesis of the Delta Plan, proposing storm surge barriers and raising coastal walls to delta height.



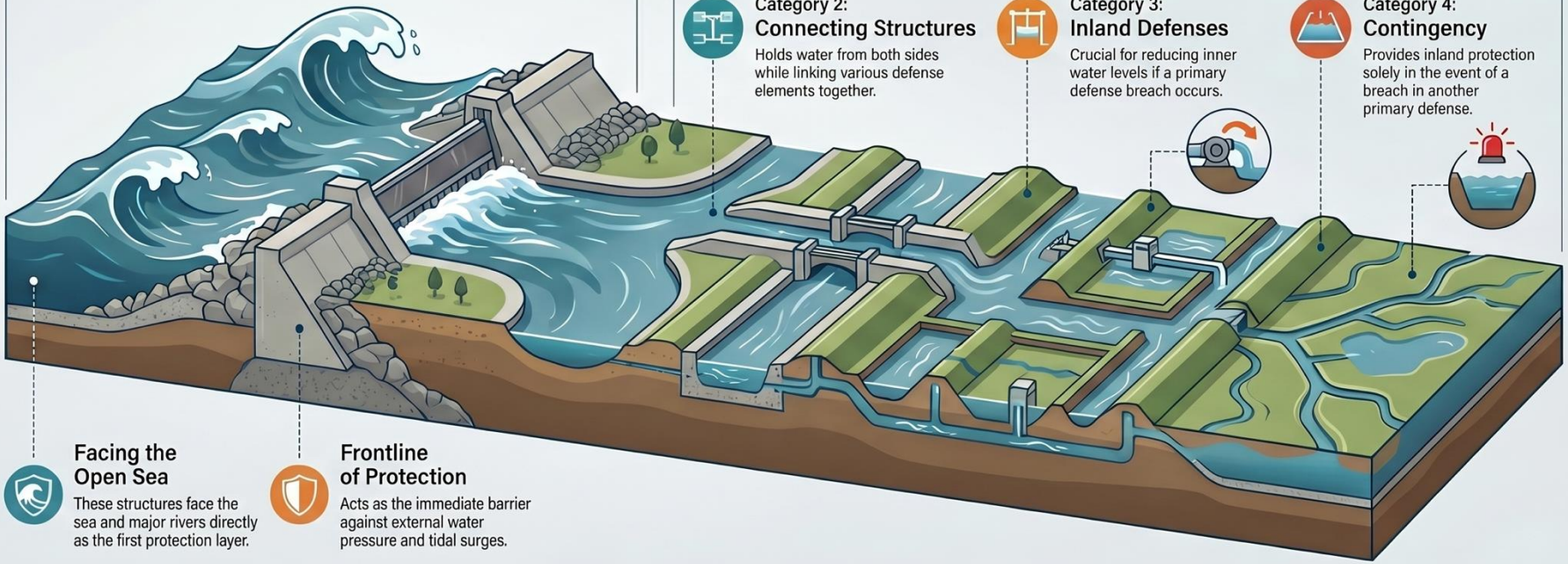

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# Anatomy of a Dutch Water System

This infographic illustrates the tiered, multi-layered hydraulic engineering strategy used to protect land from sea and river levels. It moves from the high-impact Frontline to the multi-layered Secondary Defenses.

## Primary Defenses

## Secondary Defenses



**Category 2:**
**Connecting Structures**
Holds water from both sides while linking various defense elements together.

**Category 3:**
**Inland Defenses**
Crucial for reducing inner water levels if a primary defense breach occurs.

**Category 4:**
**Contingency**
Provides inland protection solely in the event of a breach in another primary defense.

**Facing the Open Sea**
These structures face the sea and major rivers directly as the first protection layer.

**Frontline of Protection**
Acts as the immediate barrier against external water pressure and tidal surges.



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B

# What is a defence ?

# This is a dike !



(Photo https://www.destadgorinchem.nl/lokaal/natuur-en-milieu/603483/ook-drie-droge-zomers-deren-de-dijken-niet)

# This is a (coastal) dike as well !



(https://zeestad.nl/nieuws/onderzoek-naar-wonen-op-onbreekbare-dijk-helder)

# This is a breakwater !



Breakwater of the port of Ericeira – Lisbon - Portugal.

# This is a tragedy !



Dike Breach in Huarong County - China (2016)

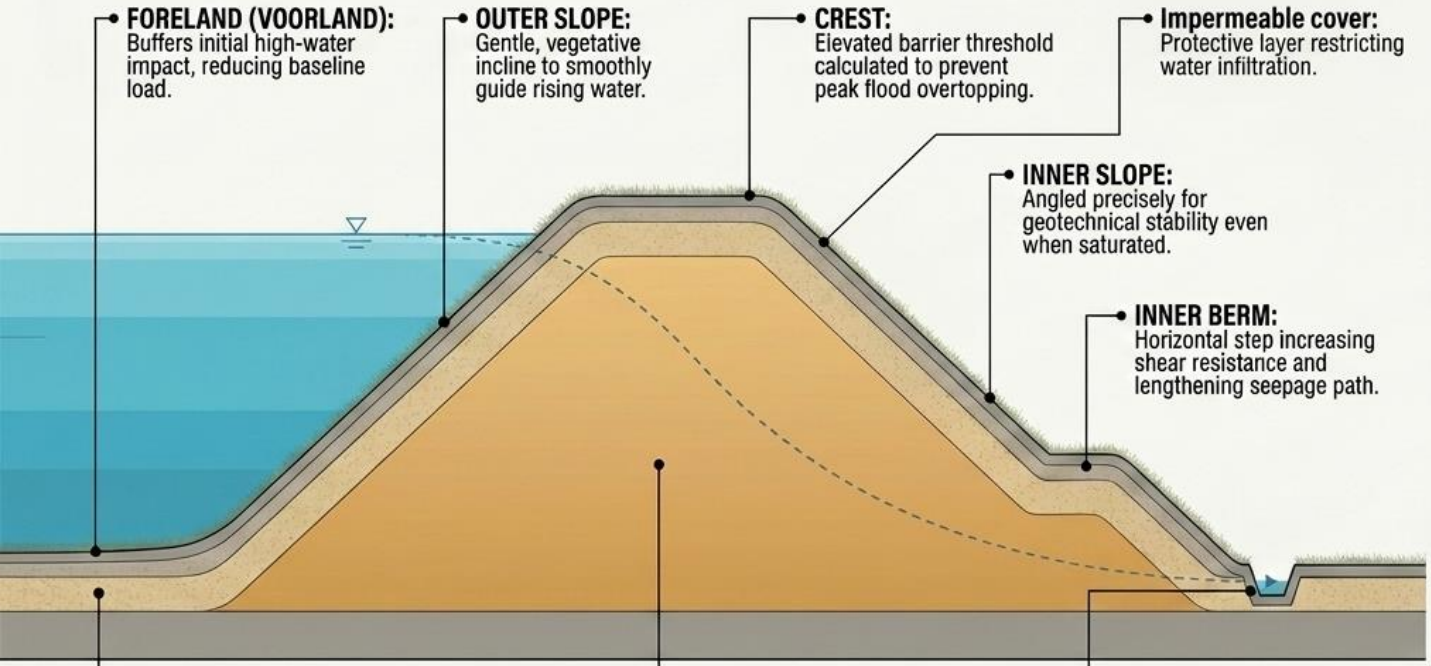
https://www.chinadaily.com.cn/china/2016-07/10/content_26030972_3.htm




Sea Wall failed , Portsmouth UK (2015)

https://www.bbc.com/news/uk-england-hampshire-35289763

# FLUVIAL DEFENSE | THE RIVER DIKE

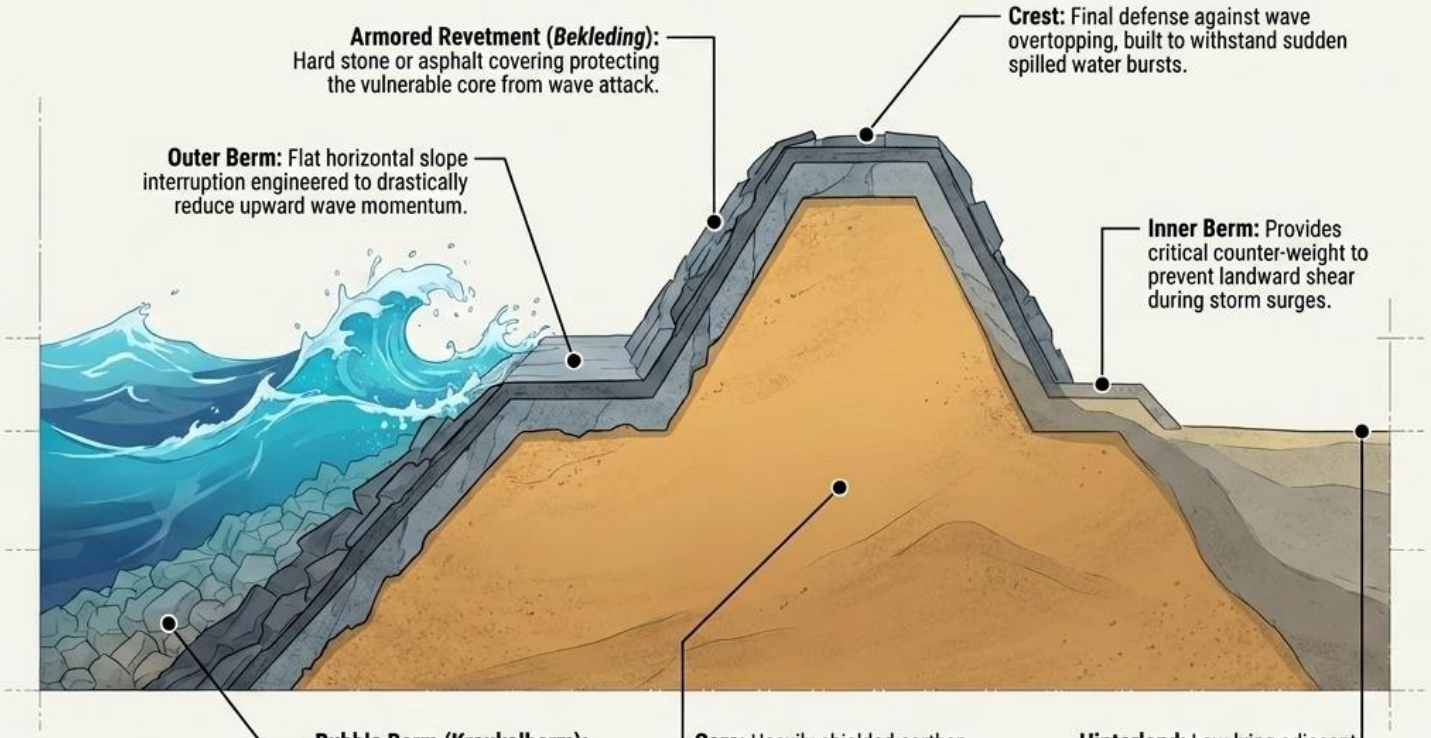



## DESIGN PROFILE

| | |
|---|---|
| **ENVIRONMENT:** | INLAND / FLUVIAL |
| **PRIMARY THREAT:** | SEEPAGE, PIPING & PROLONGED HIGH WATER |
| **CORE MATERIAL:** | EARTHEN (SAND & CLAY) |
| **ENERGY MANAGEMENT:** | GUIDING GENTLE FLOWS AND MANAGING HYDROSTATIC PRESSURE |

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# COASTAL DEFENCE | THE SEA DIKE




## DESIGN PROFILE

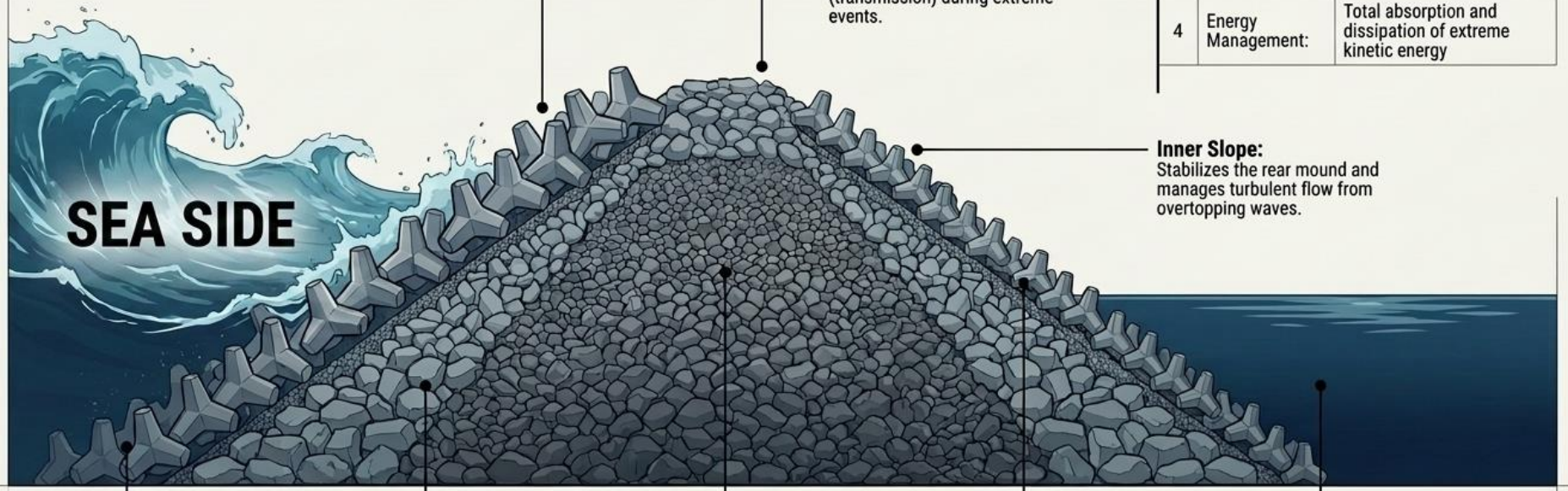
| | |
|---|---|
| **Environment:** | Coastal / Littoral |
| **Primary Threat:** | Extreme Wave Run-up & Outer Erosion |
| **Core Material:** | Composite (Earth Base + Hard Armor Revetment) |
| **Energy Management:** | Breaking wave momentum and resisting kinetic impact |

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# OFFSHORE DEFENSE | THE RUBBLE MOUND BREAKWATER

## DESIGN PROFILE

| | | |
|---|---|---|
| 1 | Environment: | Open Sea / Offshore |
| 2 | Primary Threat: | Catastrophic Wave Impact & Structural Washout |
| 3 | Core Material: | Purely Artificial (Graded Quarry Rock & Concrete) |
| 4 | Energy Management: | Total absorption and dissipation of extreme kinetic energy |




**Armor Units (Tetrapods):**
Massive interlocking concrete blocks absorbing catastrophic breaking wave energy.

**Crest:**
Intentionally designed to allow calculated wave overtopping (transmission) during extreme events.

**Inner Slope:**
Stabilizes the rear mound and manages turbulent flow from overtopping waves.

**Armor Units (Tetrapods):**
Massive interlocking concrete blocks absorbing catastrophic breaking wave energy.

**Filter Layer:**
Graded intermediate zone locking finer core materials in place against washout.

**Impermeable Core:**
Dense inner quarry rock regulating internal permeability and preventing pressure buildup.

**Inner Slope:**
Stabilizes the rear mound and manages turbulent flow from overtopping waves.

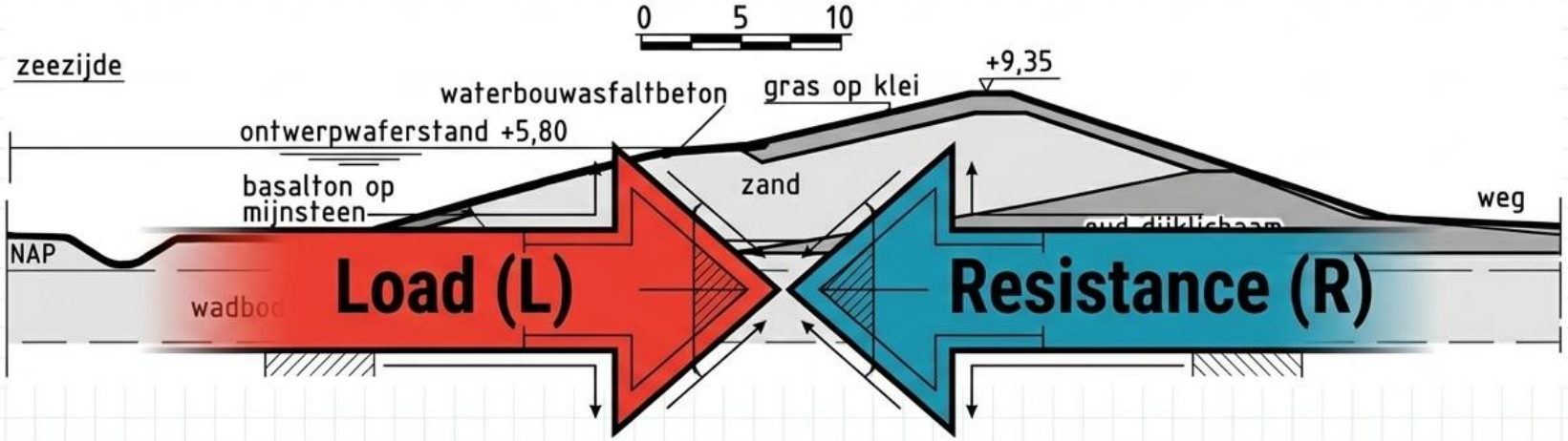
**Lee Side:**
Engineered zone of dissipated energy, creating safe and calm harbor conditions.

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C

# How does it fail ?

# The Limit State Equation: **Z = R - L**

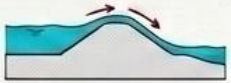



**R = Resistance** (Material strength, geometry, and weight of the dike)
**L = Load** (Water pressure, wave impact, gravity, pore pressure)
**Z = The Safety Margin**

**The Engineering Mandate:** If **Z < 0**, the system fails.
Every mechanism we study is just a different way **L** overcomes **R**.

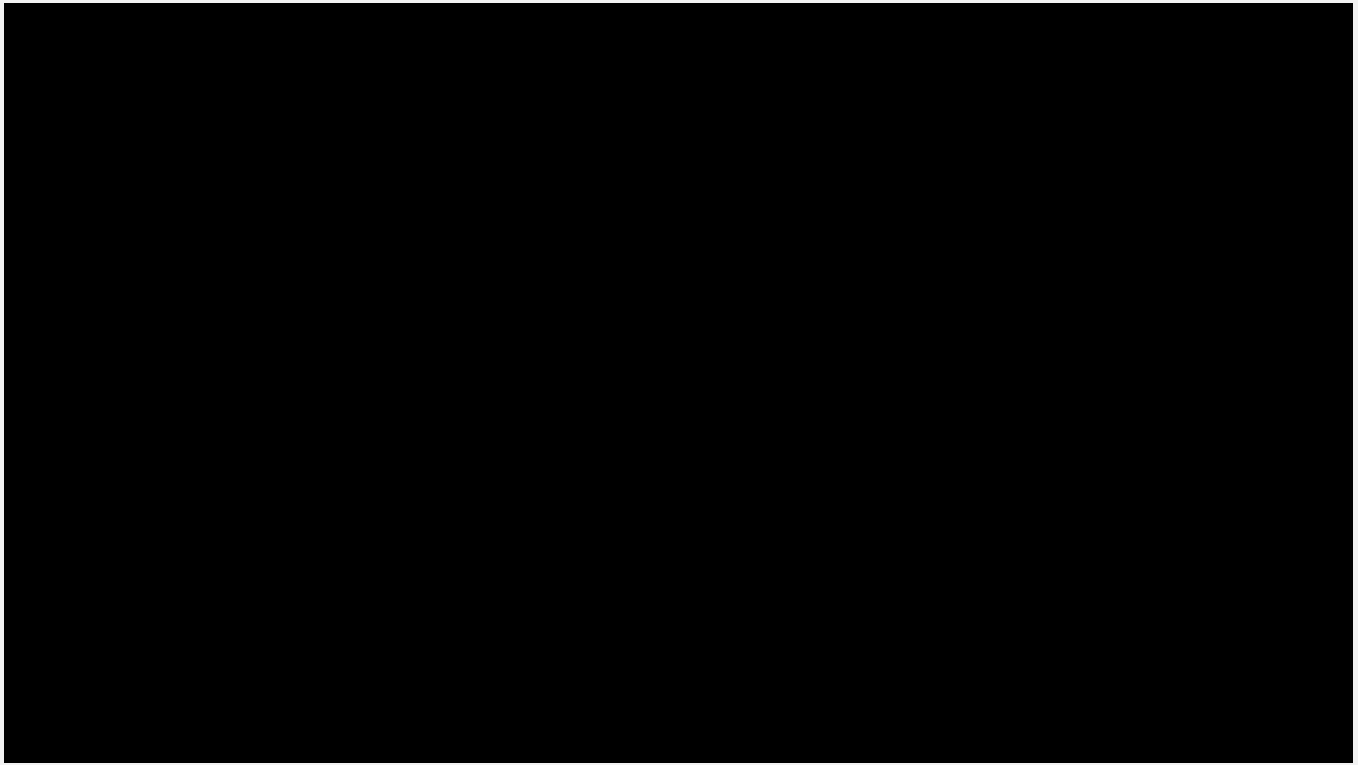
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# Dike Failure Mechanisms

| Icon | Mechanism | Limit State Vulnerability | Primary Load (L) | Primary Resistance (R) |
|---|---|---|---|---|
|  | Overflow | Crest elevation surpassed | Surge Level | Crest Height |
|  | Overtopping | Landward slope erosion | Wave Energy | Talus Quality |
|  | Macro-stability | Deep rotational slip | Weight | Friction and Cohesion |
|  | Piping | Underground void formation | Pore Pressure | Seepage Length |
|  | Settlement | Gradual height loss | Time/Compaction | Overheight |
|  | Ice Collision | Surface scraping | Ice Mass | Revetment Armor |
|  | Breakwater Failure | Armor displacement | Wave Height | Armor Weight |

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# Fishtank Experiments

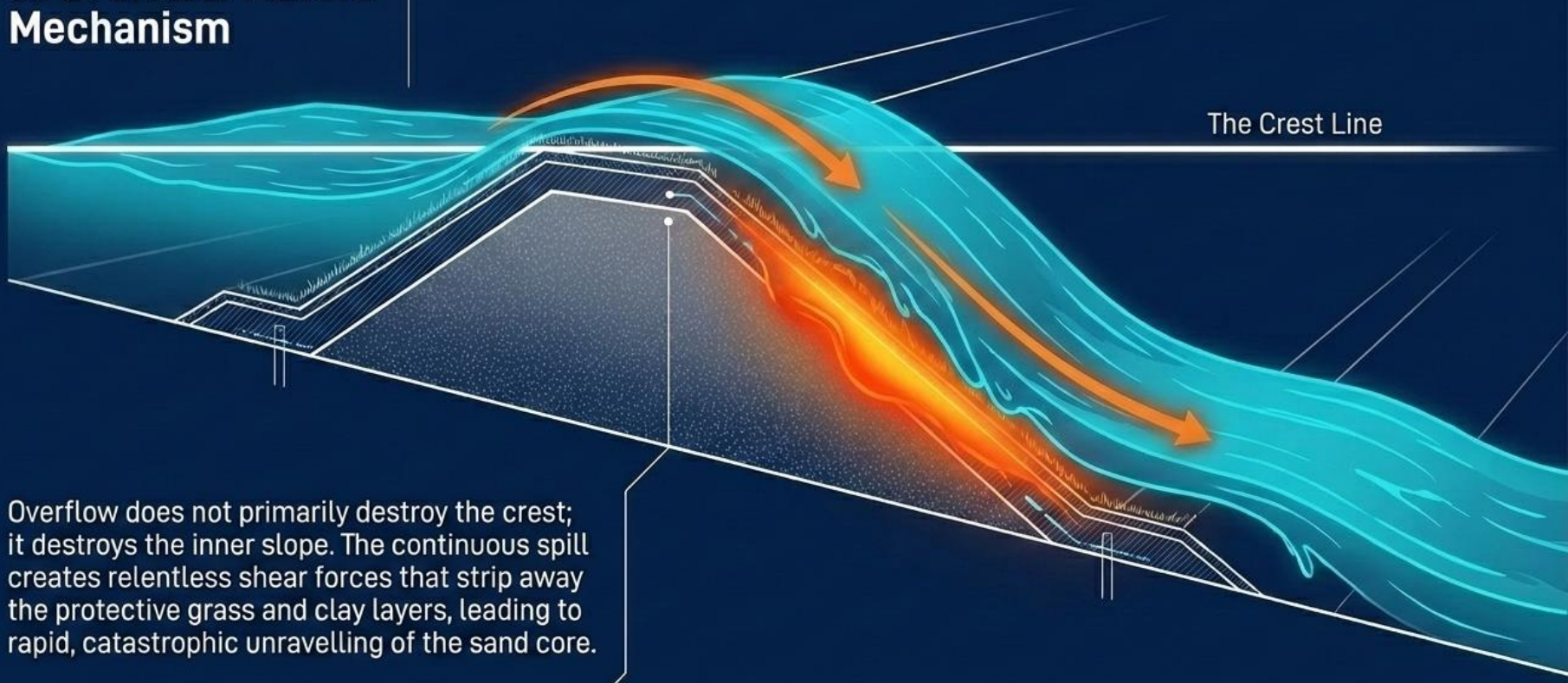


https://youtu.be/WShYyfLaCmo?si=EyPidj2WvBtnPYQU

# C.1

## Overflow

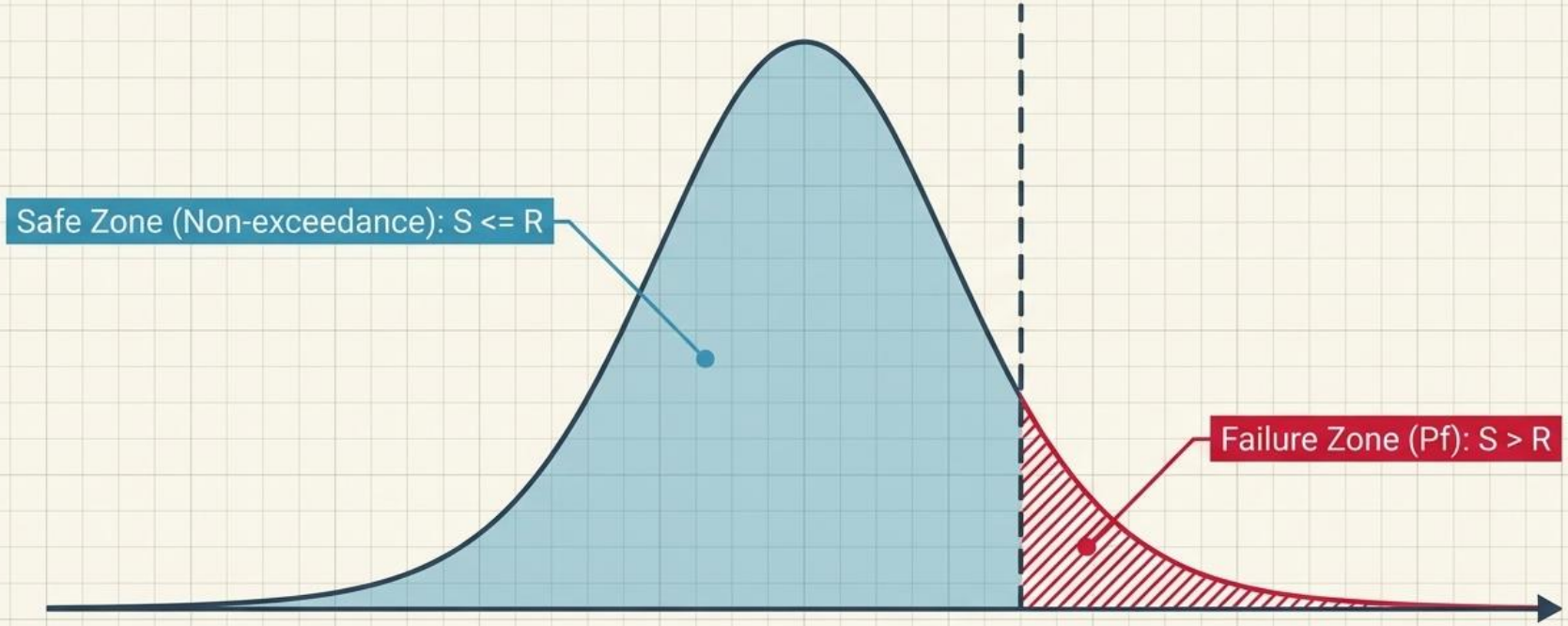
# Physical Process of Overflow Failure Mechanism




Overflow does not primarily destroy the crest; it destroys the inner slope. The continuous spill creates relentless shear forces that strip away the protective grass and clay layers, leading to rapid, catastrophic unravelling of the sand core.

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# Quantifying Risk: Probability of Failure (Pf)

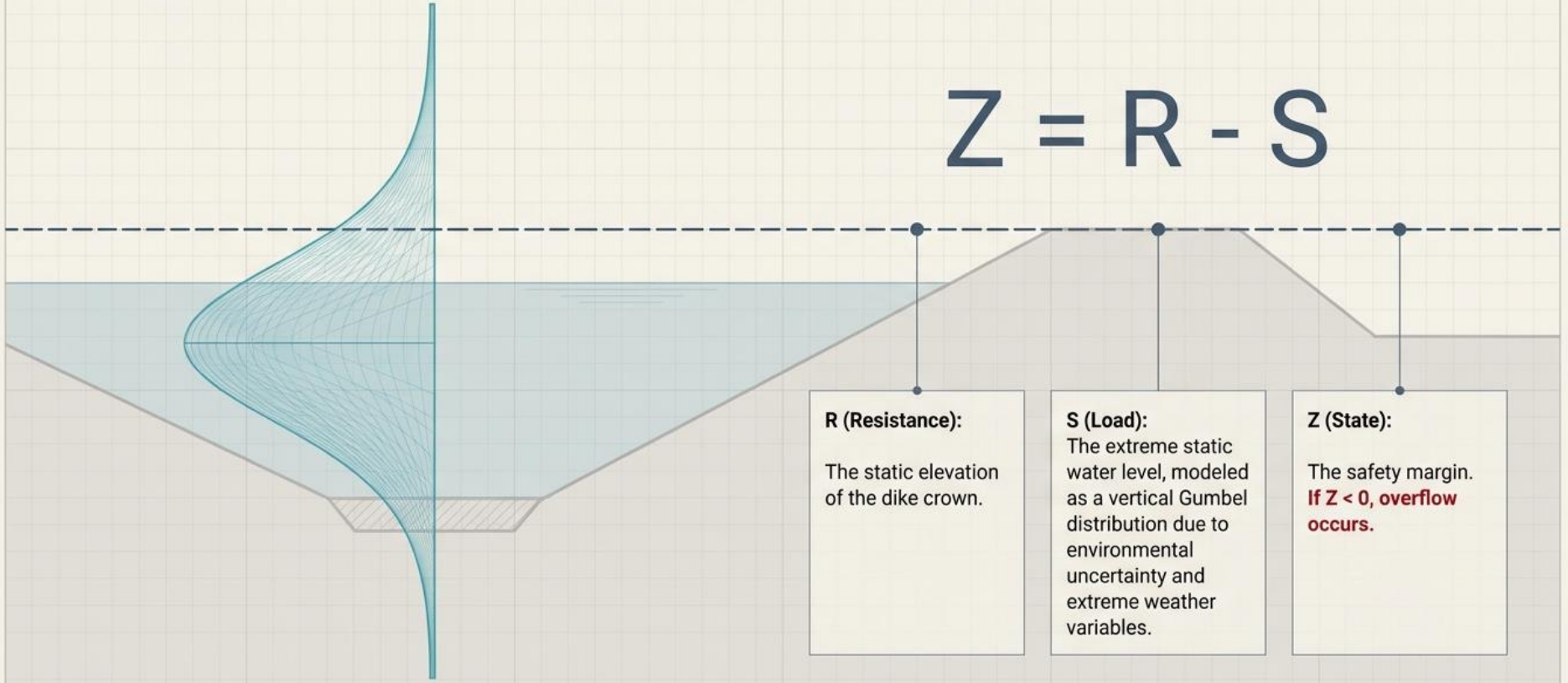



The Probability of Failure (Pf) is not a single point, but a calculated area. It is the exact mathematical integral of the distribution where the environmental Load (S) overtakes the structural Resistance (R).

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# Modeling Uncertainty: The Limit State Equation

$$Z = R - S$$



**R (Resistance):**

The static elevation of the dike crown.

**S (Load):**
The extreme static water level, modeled as a vertical Gumbel distribution due to environmental uncertainty and extreme weather variables.

**Z (State):**

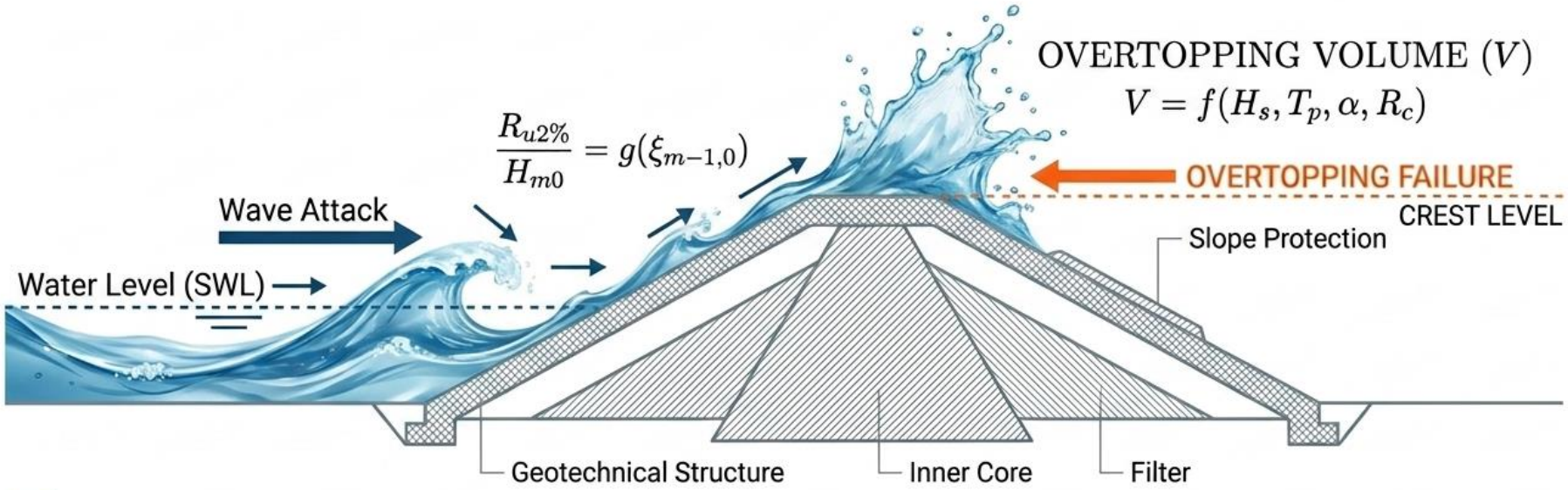
The safety margin. **If Z < 0, overflow occurs.**

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# C.2

# Wave Run-up and Overtopping

# DIKE FAILURE MECHANISMS: WAVE RUN-UP AND OVERTOPPING




Deep Blue = Hydrodynamics
Slate Gray = Geotechnics
Alert Orange = Failure Limits

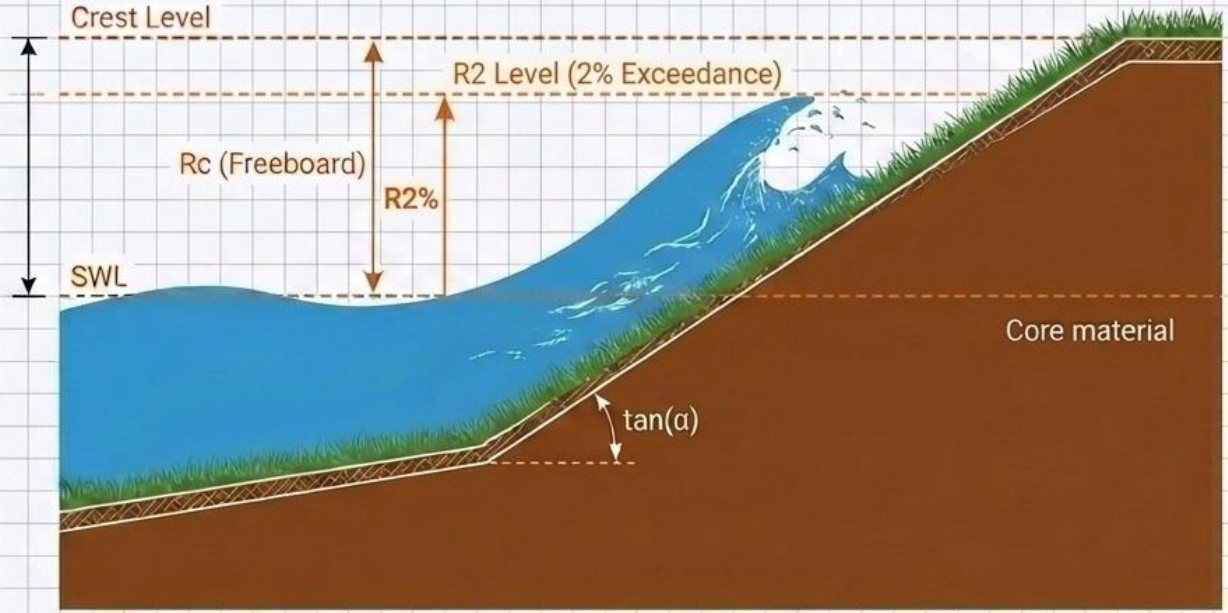
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# Wave run up and the $R_{2\%}$

## 1 Core Concept

**Rc (Freeboard):** The structure's crest level relative to the Still Water Level (SWL).

**R2 (2% Exceedance Run-Up):** The vertical height of wave run-up above SWL that is exceeded by only 2% of incoming waves. (Note: R2 < Rc for no overtopping).




**Insight: Overtopping only occurs when the R2 wave run-up level surpasses the Rc freeboard level (R2 > Rc).**

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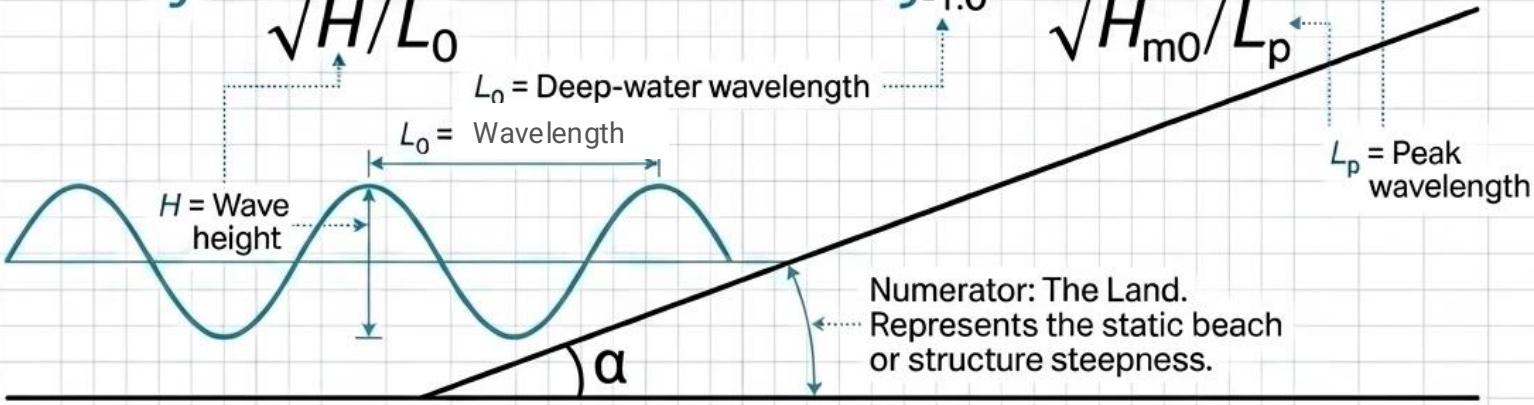
# Breaker Parameter (Iribarren number)

**Average Iribarren Number (ξ)**

$$\xi = \frac{\tan(\alpha)}{\sqrt{H/L_0}}$$

**First Moment Spectral Iribarren Number ($\xi_{-1.0}$)**

$$\xi_{-1.0} = \frac{\tan(\alpha)}{\sqrt{H_{m0}/L_p}}$$




Deep-water wavelength:

$$L_0 = \frac{g}{2\pi} \cdot T^2$$

**ξ is fundamentally a ratio of slope steepness to wave steepness.**

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# The Energy Transfer Continuum

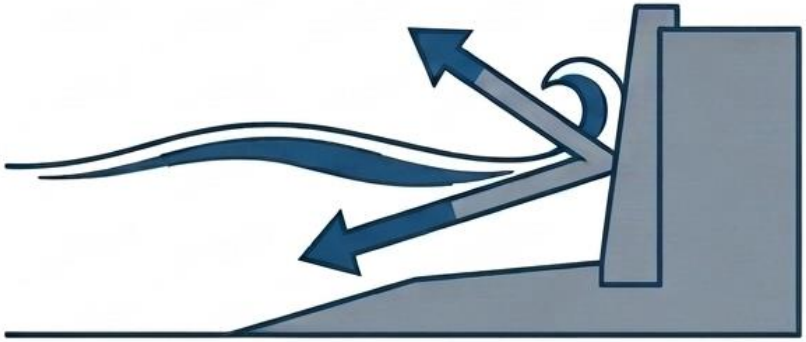
## Low $\xi_{m-1,0}$



## Turbulence & Dissipation

Lower Iribarren numbers (gentle slopes, steeper waves) force the wave to break, dissipating its energy through turbulence before reaching the upper shore.

## High $\xi_{m-1,0}$

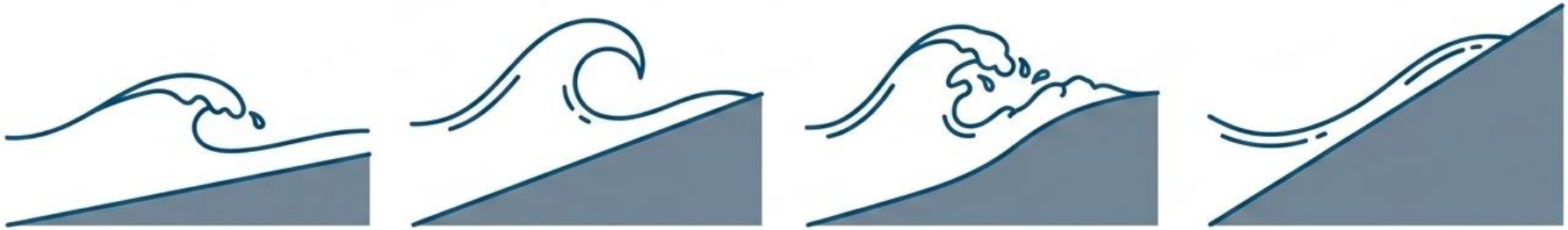


## Surging & Reflection

Higher Iribarren numbers (steep slopes, flatter waves) prevent breaking. The wave's energy is highly **reflected** back into the ocean, heavily impacting coastal structures.

The Surf Similarity Parameter acts as the coastal blueprint—predicting **whether a shoreline will absorb** ocean energy or reflect it.

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**Spilling**

Gentle slope.
Long life span.

**Plunging**

Strong turbulence
and dissipation.

**Collapsing**

The critical transition
point (overgang).

**Surging**

Little breaking,
high reflection.

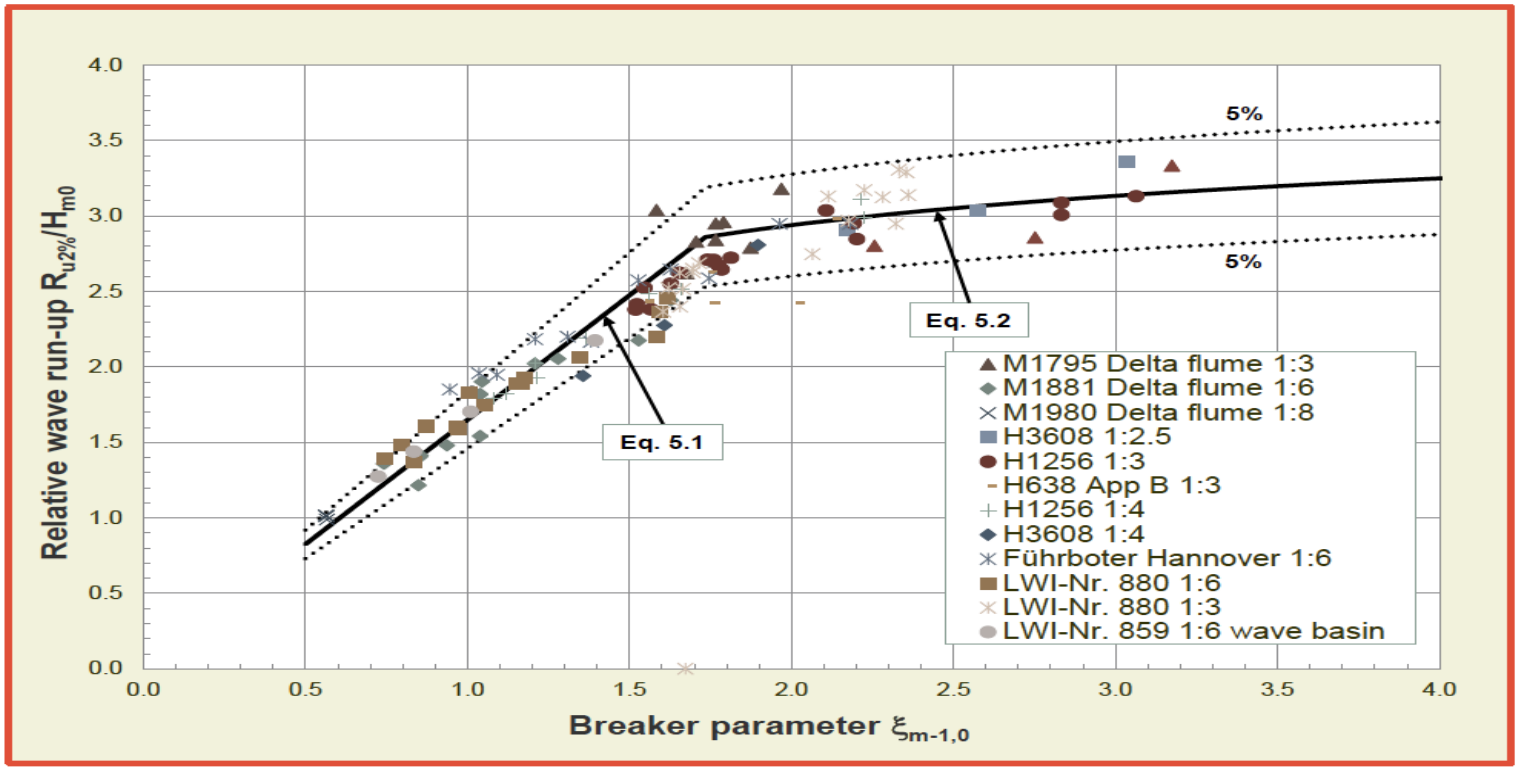
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| Breaker Type | $\xi_{m-1,0}$ Range | Beach Profile | Physical Characteristics |
|---|---|---|---|
| Spilling | No specific numeric limit | Mostly Horizontal | Relatively long life span, surface foam |
| Plunging | $0.5 < \xi_{m-1,0} < 1.8$ | Moderately Steep | Curling crest over an air pocket; strong turbulence and dissipation |
| Collapsing | Transition | Transition | Overgang (Transition state) |
| Surging | $\xi_{m-1,0} > 1.8$ | Very Steep | Little breaking, high wave reflection |

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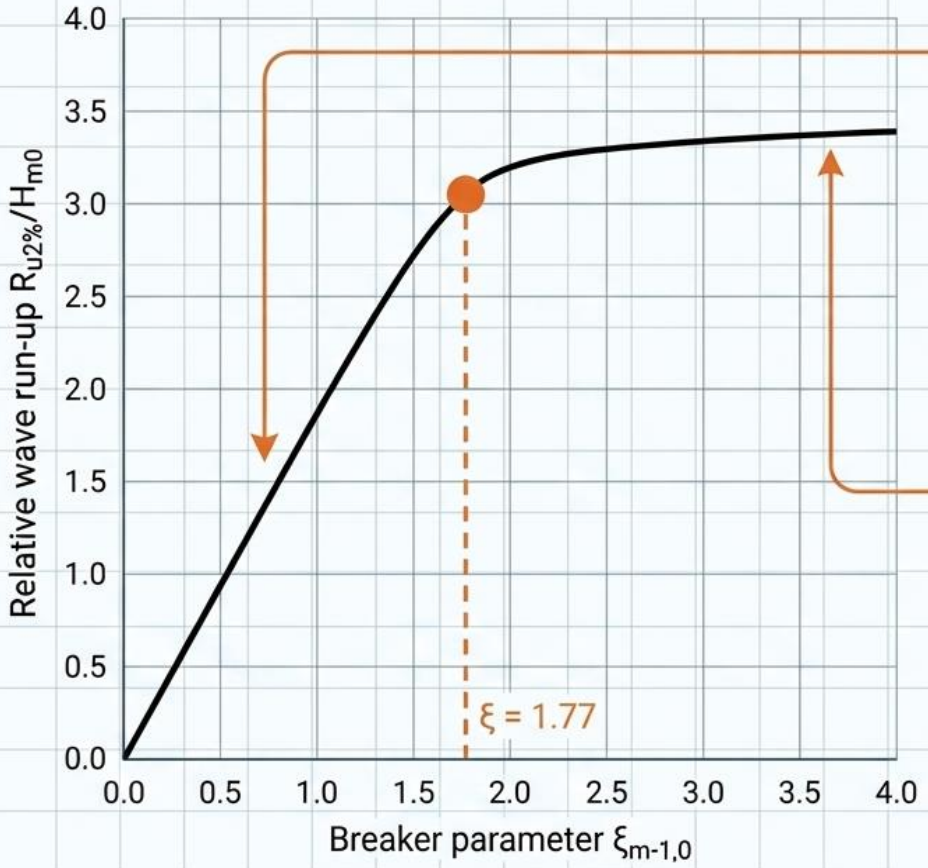
# Empirical flume data dictates European design limits.

Relative wave run-up scales linearly during plunging waves, but caps logarithmically as waves transition to surging behavior at $\xi > 1.77$.




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# Theoretical Wave Run-up $R_{U2\%}$




For Plunging Waves ($\xi_{m-1,0} \leq 1.77$)

$$\frac{R_{u2\%}}{H_{m0}} = 1.75 \cdot \xi_{m-1,0}$$

For Surging Waves ($\xi_{m-1,0} > 1.77$)

$$\frac{R_{u2\%}}{H_{m0}} = 4.3 - \frac{1.6}{\sqrt{\xi_{m-1,0}}}$$

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# Actual $Ru_2\%$ of composite front slopes

## Plunging Waves ($\xi_{m-1,0} \leq 1.77$)

$$\frac{R_{u2\%}}{H_{m0}} = 1.75 \cdot \gamma_b \cdot \gamma_f \cdot \gamma_\beta \cdot \xi_{m-1,0}$$

### Coefficient Effects:

- **$R_{u2}\%/H_{m0}$:** Relative 2% run-up height.
- **$\gamma_b$ (Berm):** Reduction for berms (breaks energy).
- **$\gamma_f$ (Roughness):** Reduction for surface friction.
- **$\gamma_\beta$ (Oblique):** Reduction for angle of attack.

⚠ **Theoretical Safety Limit:** The multiplication of the 3 factors ($\gamma_b \cdot \gamma_f \cdot \gamma_\beta$) must be greater than 0.4.

## Surging Waves ($\xi_{m-1,0} > 1.77$)

$$\frac{R_{u2\%}}{H_{m0}} = \gamma_b \cdot \gamma_f \cdot \gamma_\beta \cdot \left(4.3 - \frac{1.6}{\sqrt{\xi_{m-1,0}}}\right)$$

### Coefficient Effects:

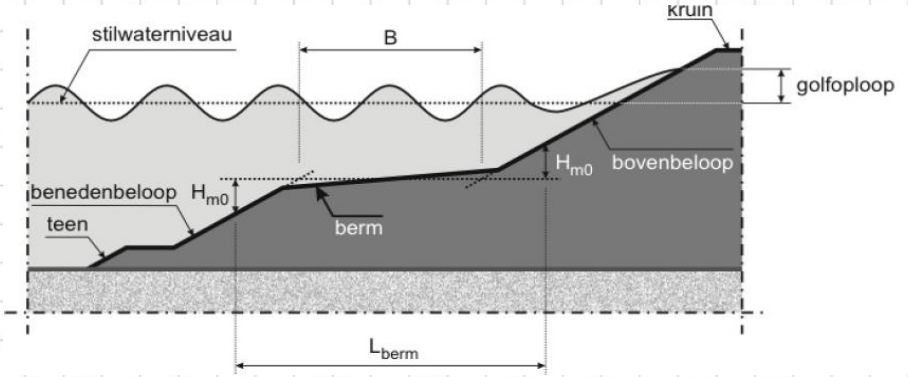
- **$R_{u2}\%/H_{m0}$:** Relative 2% run-up height.
- **$\gamma_b$, $\gamma_f$, $\gamma_\beta$:** Reduction factors for berm, roughness, and oblique waves (as above).
- **$\xi_{m-1,0}$ (Breaker Param.):** Slope/wave steepness

⚠ **Theoretical Safety Limit:** The multiplication of the 3 factors ($\gamma_b \cdot \gamma_f \cdot \gamma_\beta$) must be greater than 0.4.

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# Correcting for geometric berms ($\gamma_b$)

The berm influence factor ($\gamma_b$) is calculated based on the berm width ($B$), berm length ($L_{\text{berm}}$), water depth over the berm ($d_b$), run-up height ($R_{u2\%}$), and wave height ($H_{m0}$).




## Estimating Berm Influence ($\gamma_b$)

$$\gamma_{b_\max} = 1 - \frac{B}{L_{\text{berm}}} * (1 - r_{dB})$$

$$r_{dB} = 0.5 - 0.5 * \cos\left(\frac{\pi * d_b}{R_{u2}\%}\right) \quad (\text{for } d_b > 0, \text{Submerged})$$

$$r_{dB} = 0.5 - 0.5 * \cos\left(\frac{\pi * d_b}{2 * H_{m0}}\right) \quad (\text{for } d_b \leq 0, \text{Emerged})$$

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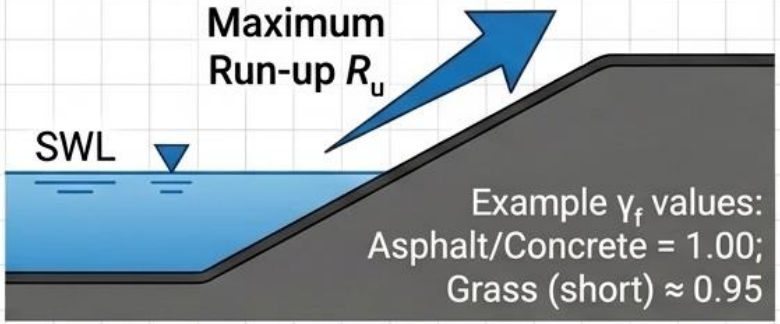
# Correcting for revetment friction ($\gamma_f$).

Rougher materials increase hydraulic friction, dampening wave run-up. The correction factor is a weighted average of materials **within the wave's zone of influence** (0.25 Ru below to 0.50 Ru above SWL).

$$\gamma_f = \frac{\sum L_i * \gamma_{f,i}}{\sum L_i}$$

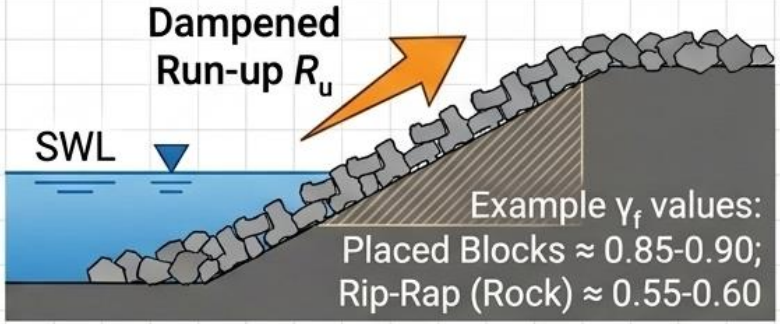
Weighted average based on segment lengths ($L_i$) and friction factors ($\gamma_{f,i}$).

## Smooth Surface (e.g., Asphalt)




$\gamma_f$ = 1.0. Low friction.

## Rough Surface (e.g., Rip-Rap)

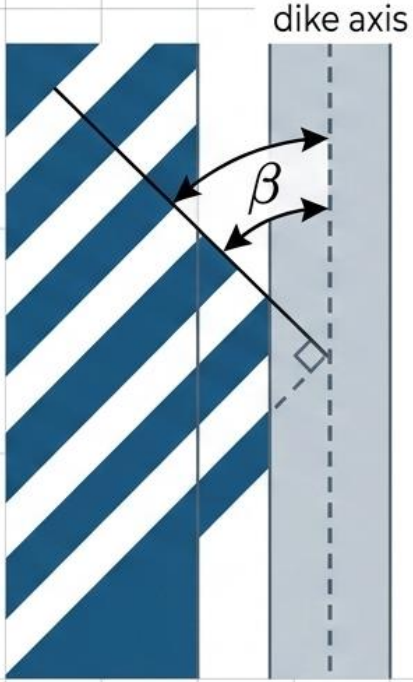



$\gamma_f$ < 1.0. High friction.

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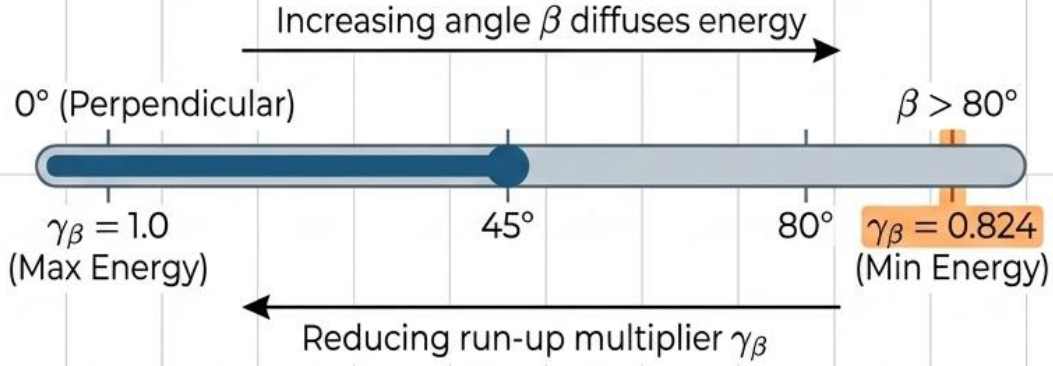
# Correcting for wave incidence direction ($\gamma_\beta$).

Waves rarely strike perpendicular to the dike. Oblique incidence diffuses kinetic energy parallel to the crest, mathematically dampening the run-up multiplier.




$$\gamma_\beta = 1 - 0.0022|\beta| \quad (\text{for } 0° \leq |\beta| \leq 80°)$$

$$\gamma_\beta = 0.824 \quad (\text{for } |\beta| > 80°)$$

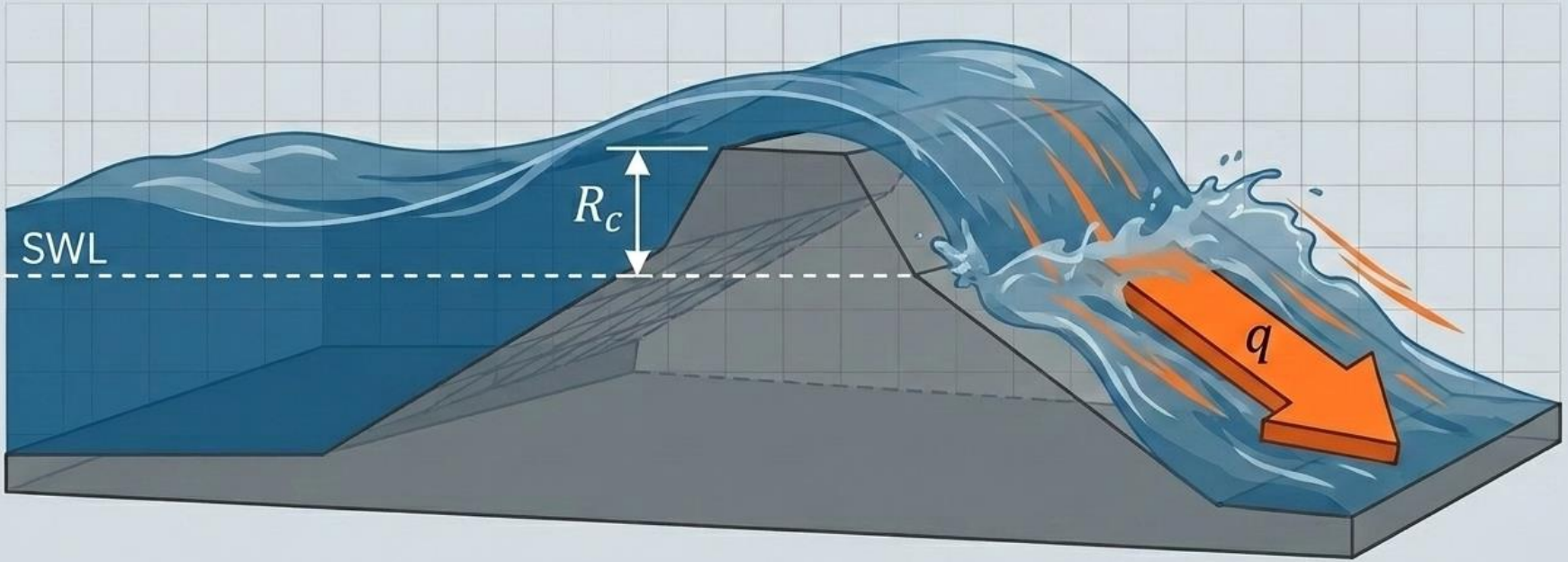



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# Wave Overtopping

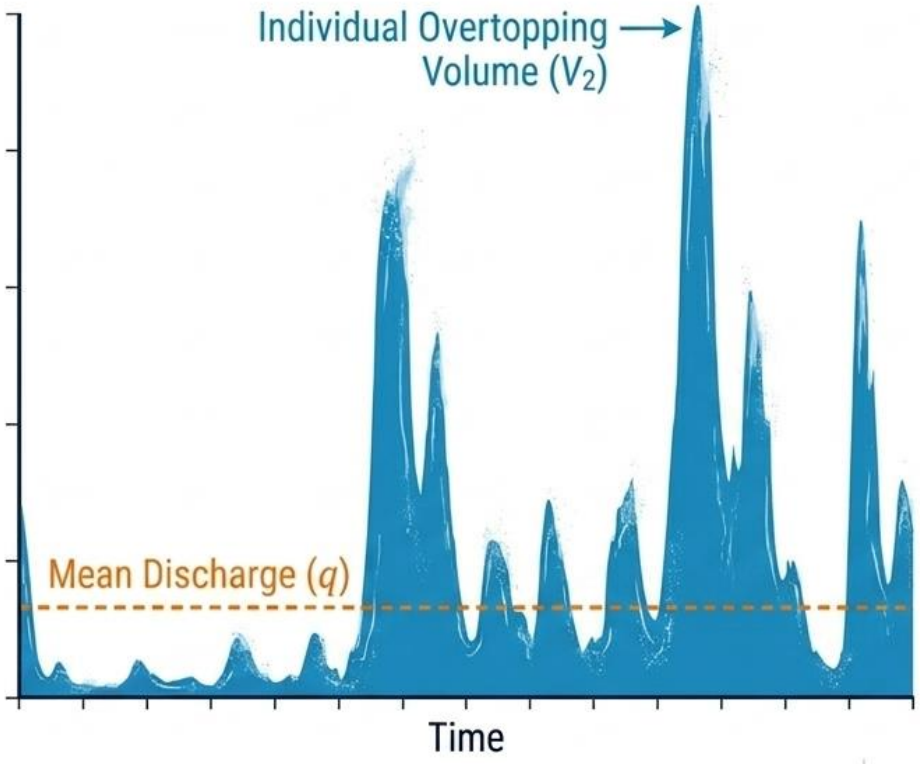
If $R_{u2\%}$ exceeds the available freeboard ($R_c$), the framework shifts from measuring vertical height (meters) to calculating destructive volumetric discharge ($q$, measured in liters per second per meter of dike width).

$R_c$ (Freeboard) $q$ (Discharge)




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# Averages Hide Monsters: Mean vs. Extreme Volumes




Mean overtopping discharge ($q$) is averaged over thousands of waves. However, structural failure is often caused by the 2% largest individual overtopping volumes ($V_2$), which follow a log-normal distribution.

## The Infragravity Catalyst

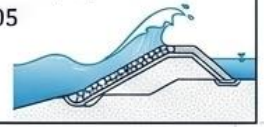
The largest volumes in shallow water occur when short waves "ride" on the crests of low-frequency infragravity waves.

### Extreme Volume Formula ($V_2$):

$$V_2 = 5.56 \cdot H^2 \cdot \exp\left(-9.94 \cdot \frac{R_c \cdot {s_{HF}}^{0.57}}{\gamma_f \cdot H}\right)$$

Coupled with extreme overtopping velocities ($u_2$):

$$u_2 = 1.26 \cdot \sqrt{gh} \cdot \left[\frac{R_c \cdot {s_{HF}}^{-0.24}}{\gamma_f \cdot H}\right]^{-1.05}$$



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Based on Slide C

# Calculating Overtopping Discharge (q) & Maximum Discharge ($q_{max}$)

Because freeboard sits inside a negative exponential function, adding even minor height to the dike crest results in a massive, non-linear reduction in the total volume of water flooding the protected lowland.

$$A = \pi r^2$$

**Discharge (q)**

$$\frac{q}{\sqrt{gH_{m0}^3}} = 0.067\sqrt{\tan(\alpha\gamma_b\xi_{m-1,0}}\exp\left(\frac{-4.3R_c}{\xi_{m-1,0}H_{\mathrm{m0}}\gamma_b\gamma_f\gamma_\beta\gamma_v}\right)$$

**Maximum Discharge ($q_{max}$)**

$$\frac{q_{\max}}{\sqrt{gH_{m0}^3}} = 0.2\exp\left(\frac{-2.3R_c}{\xi_{m-1,0}H_{\mathrm{m0}}\gamma_f\gamma_\beta}\right)$$

## $\gamma_f$ values for Crest Walls

For vertical crest walls, $\gamma_f$ represents the wall's roughness:

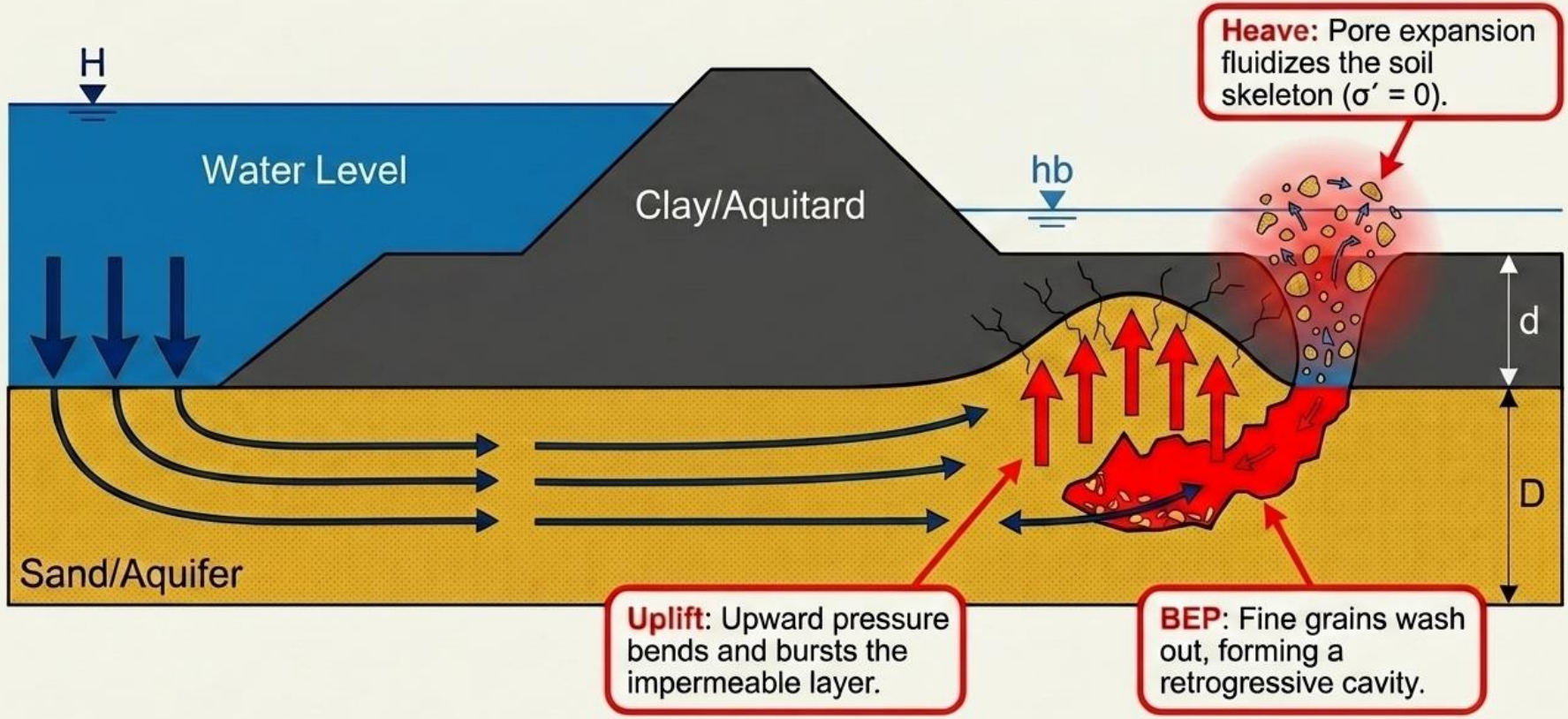
- Smooth Concrete Wall: $\gamma_f = 1.0$
- Rough Rock/Gabion Wall: $\gamma_f \approx 0.6$

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# C.3

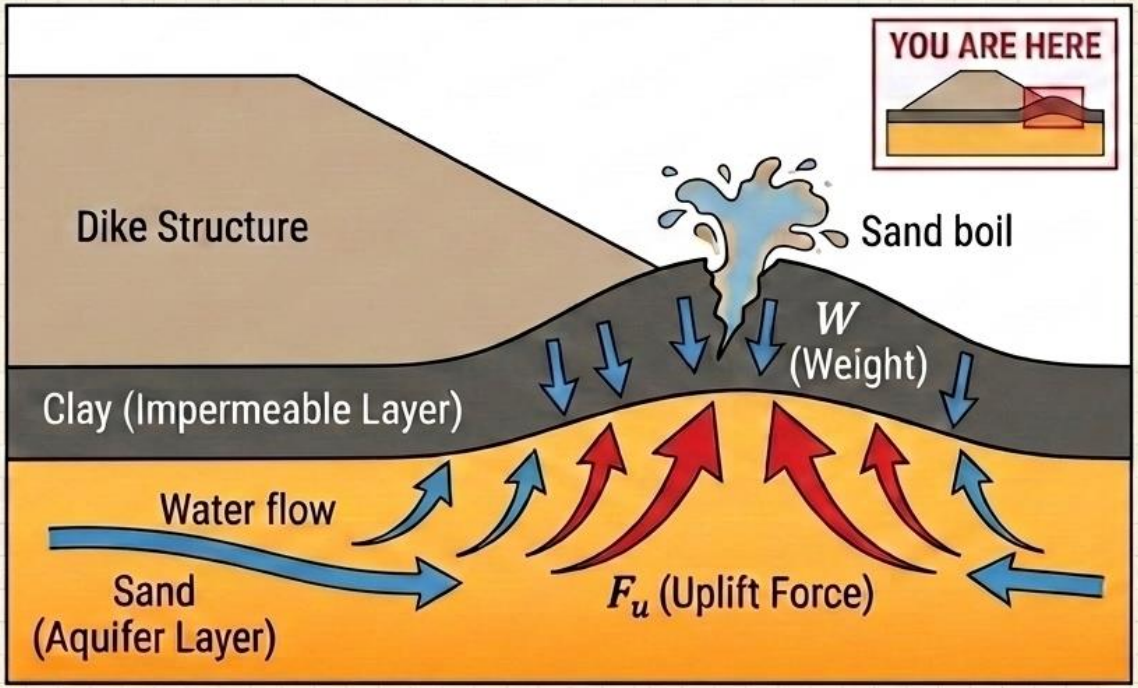
# Backward Erosion Piping (BEP)

# Physical processes of Backward Erosion Piping (BEP)




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# UPLIFT: THE PRINCIPLE OF BUOYANCY




## Formula Translation Card

**$F_u$ = Uplift Force**
Upward force due to pore water pressure ($u$) at the base of the layer.

**Equation:** $F_u = u \cdot A = h_p \cdot \gamma_w \cdot A$

**$W$ = Total Weight of Impermeable Layer**
Downward force due to saturated soil weight.

**Equation:** $W = d \cdot \gamma_{sat} \cdot A$

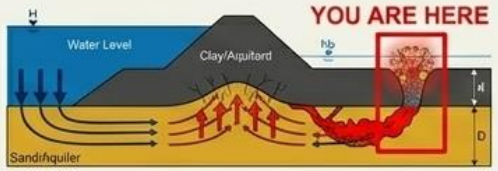
**Limit State:** $F_u = W$

(Where $A$ is area, $d$ is thickness, $h_p$ is pressure head, $\gamma_{sat}$ is saturated unit weight).

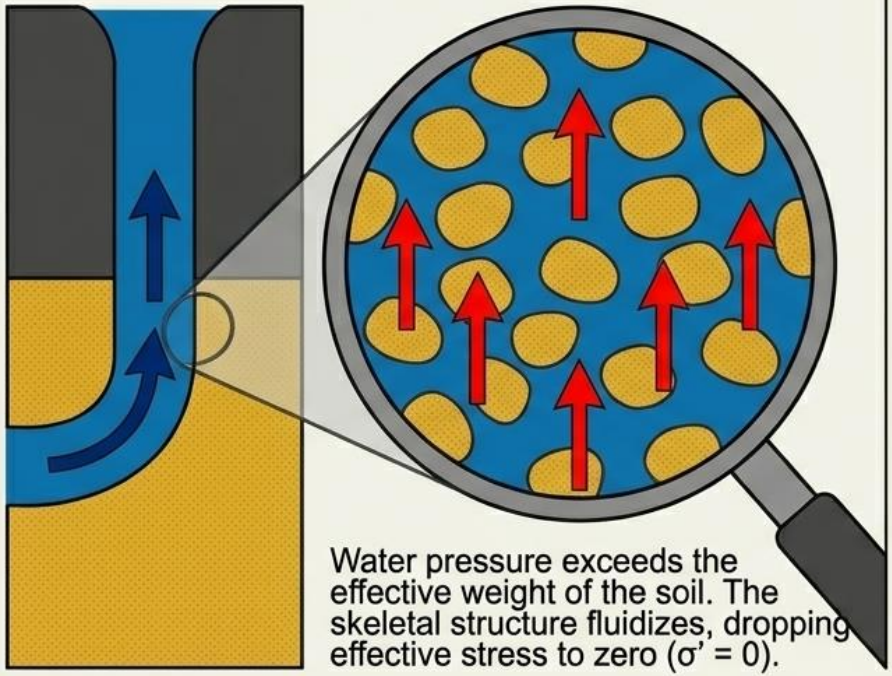
**When positive buoyancy occurs ($F_u > W$), the upward force exceeds the weight of the impermeable layer, causing it to burst and form a sand boil.**

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# HEAVE FAILURE MECHANISM & GOVERNING EQUATIONS




## Heave: The Fluidized Skeleton



Water pressure exceeds the effective weight of the soil. The skeletal structure fluidizes, dropping effective stress to zero (σ' = 0).

## Formula Translation Card

Soil porosity

$$i_{c,h} = \frac{(1-n)(\gamma_s - \gamma_w)}{\gamma_w} \approx 1.65(1-n)$$

Critical Heave Gradient (Terzaghi)

$$i = \frac{\phi_{exit} - h_p}{d}$$

Actual Exit Gradient

$$Z_h = i_{c,h} - i$$

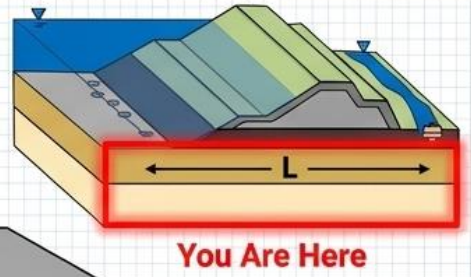
Limit State

$$FoS_h = \frac{i_{c,h}}{i}$$

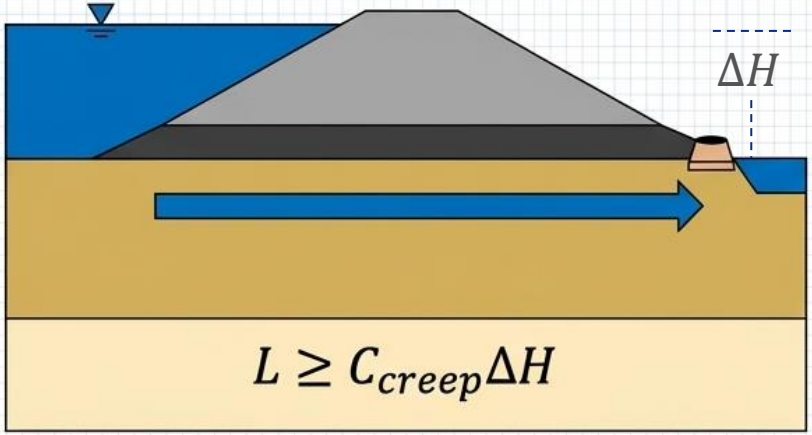
Factor of Safety

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# **Backward Erosion Piping (BEP):** Empirical Foundations

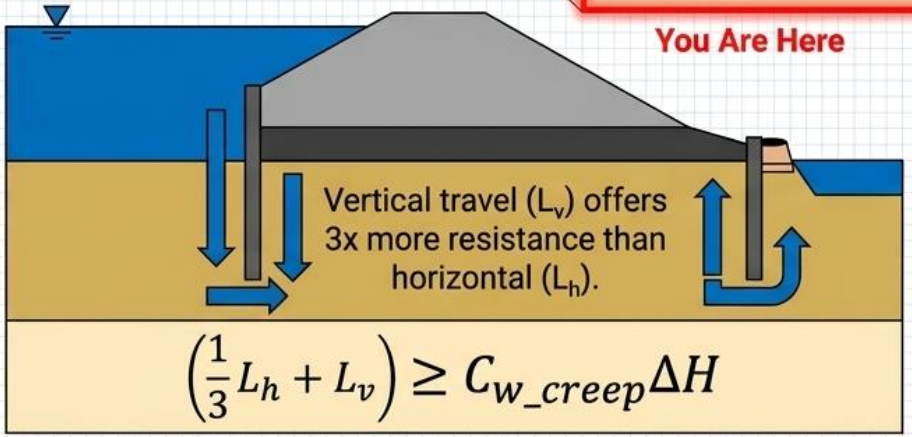



## Bligh's Rule (1910)




$$L \geq C_{creep}\Delta H$$

## Lane's Rule (1935)




$$\left(\frac{1}{3}L_h + L_v\right) \geq C_{w_creep}\Delta H$$

| Soil material | Median grain size | $C_{creep}$ (Bligh) | $C_{w_creep}$ (Lane) |
|---|---|---|---|
| Very fine sand | 63 µm to 105 µm | - | 8.5 |
| Fine sand or silt | 105 µm to 150 µm | 18 | - |
| Fine sand (micr.) | 105 µm to 150 µm | 18 | 7 |
| Coarse sand | 300 µm to 2 mm | 12 | 5 |
| Coarse gravel | 10 mm to 16 mm | - | 3 |

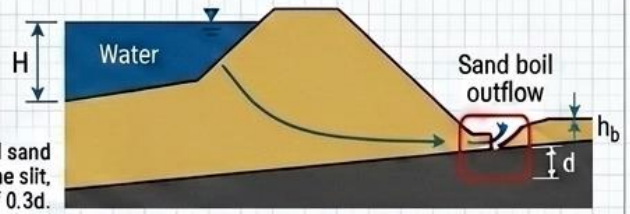
**Alert Red**

**Key Insight:** Finer soils require exponentially longer seepage paths to prevent cavity progression.

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# BEP: The Sellmeijer 2-Forces Rule (2011 & 2021)

Advanced Limit State Fluid Dynamics




**Resistance Correction:** Fluidized sand requires more energy to flow through the slit, resulting in an additional head loss of 0.3d.

$$Z_{piping} = H_c - (H - h_b - 0.3d)$$

$$H_c = m_p F_R F_S F_G L$$

## Resistance Factor ($F_R$)

$$F_R = \eta \frac{\gamma'_s}{\gamma_w} \tan(\theta)$$

- Relates to fundamental soil friction.
- 'θ' is the bedding angle of the sand grains.
- η = White's sand packing coefficient (0.25)
- $\gamma'_s$ = Unitary weight of submerged sand particles
- $\gamma_w$ = Unitary weight of water

## Scale Factor ($F_S$)

$$F_S = \frac{d_{70m}}{\sqrt[3]{\nu K L}}$$

- Relates to grain size and kinematic viscosity.
- Note calibration limit: 150μm ≤ $d_{70m}$ ≤ 430μm.
- $d_{70m}$ = Calibration reference value ($2.08 \times 10^{-4}$ m)
- ν = Kinematic viscosity of water at 20 °C
- K = Hydraulic conductivity of sand
- L = Seepage length

## Geometric Factor ($F_G$)

$$F_G = 0.91 \left(\frac{D}{L}\right)^{\frac{0.28}{(D/L)^{0.04}}}$$

- Relates to average aquifer dimensions and the specific path length.
- D = Average thickness of sand aquifer
- L = Seepage length
- Based on modelling uncertainty factor $m_p$.

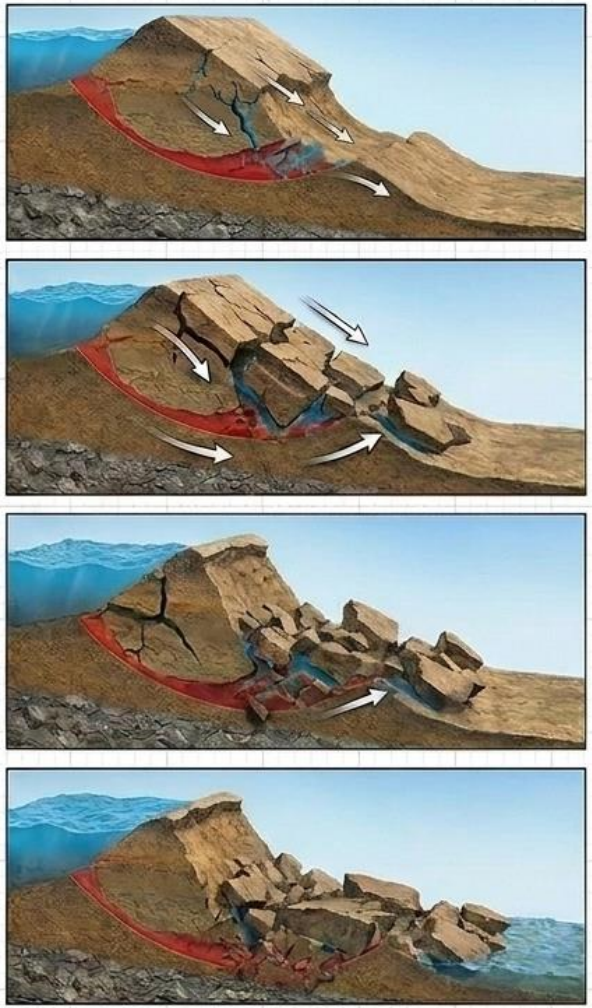
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# C.4

# Macro-stability

# MACRO-STABILITY

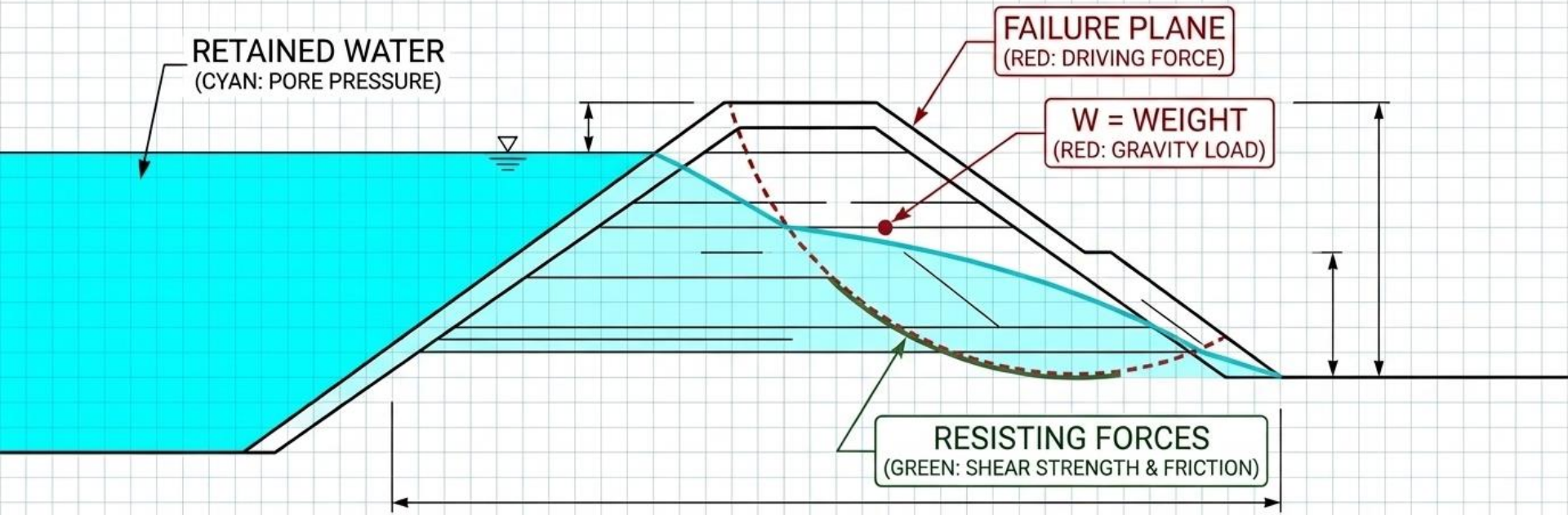


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# The Mechanics of Slip Instability

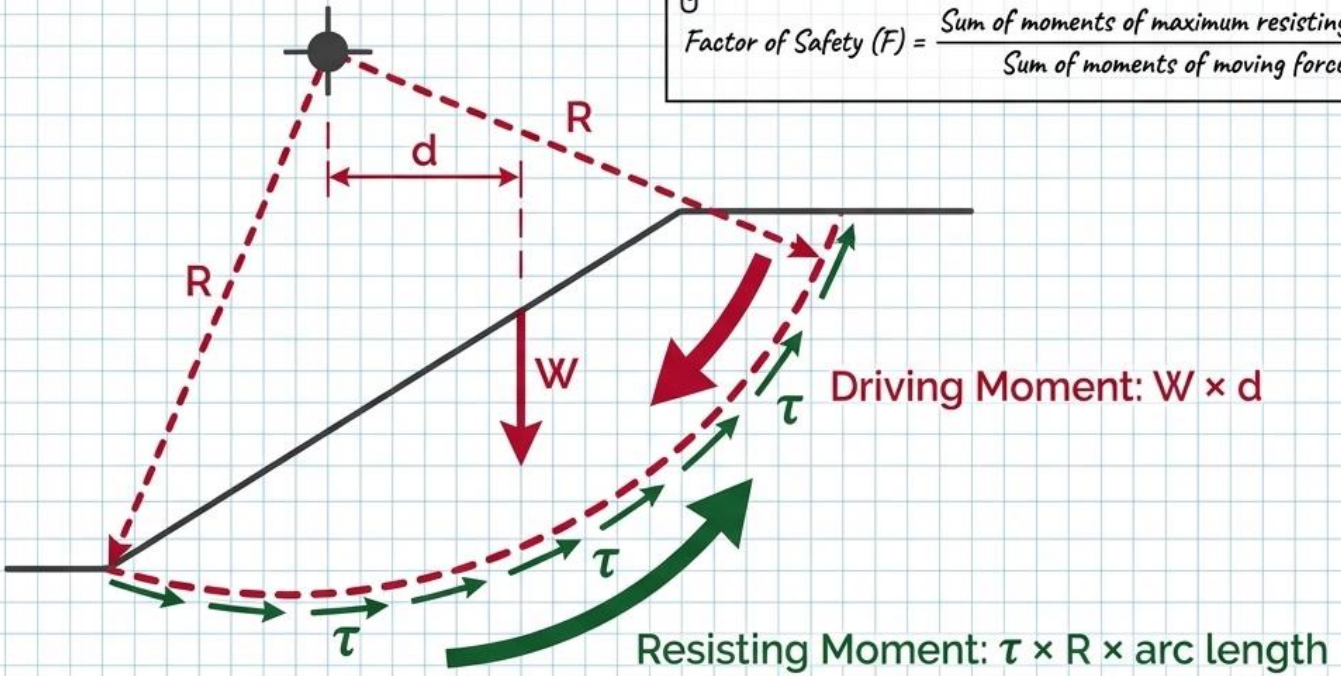
## Calculating Failure: Pore Water Pressure, Equilibrium, and Bishop's Method of Slices




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# The Anatomy of a Slip: A Strict State of Equilibrium

$$\text{Factor of Safety } (F) = \frac{\text{Sum of moments of maximum resisting forces}}{\text{Sum of moments of moving forces}}$$

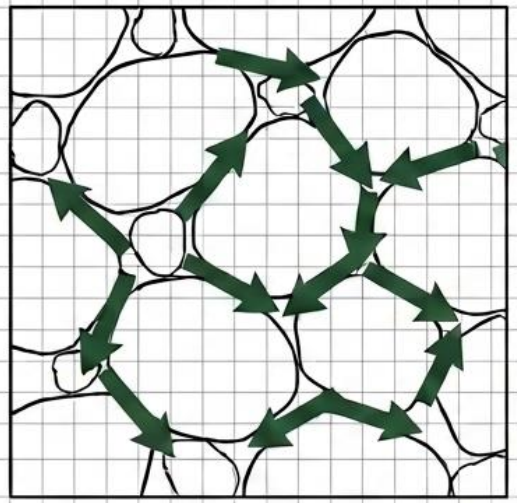



Earth structures don't just fall apart—they rotate. As long as the green resisting moment exceeds the red driving moment, the structure holds.

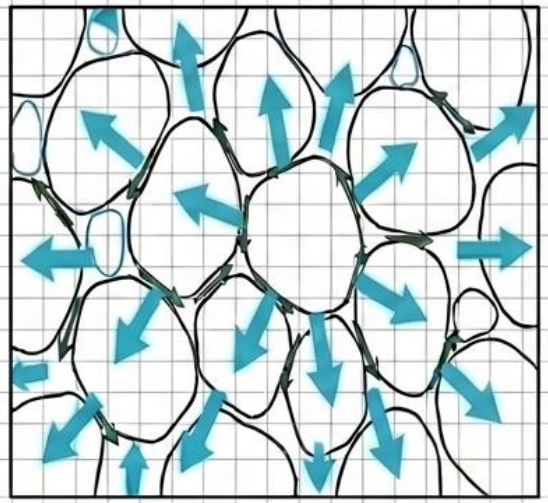
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# The Catalyst for Collapse: Pore Water Pressure

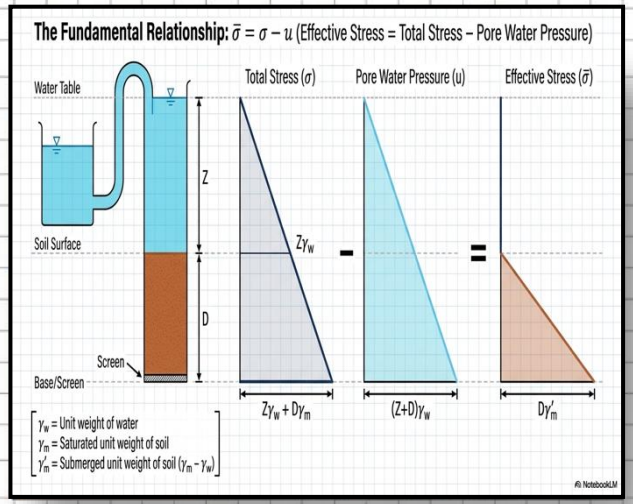
## Dry Soil



## Saturated Soil



## Phreatic Line Below Surface



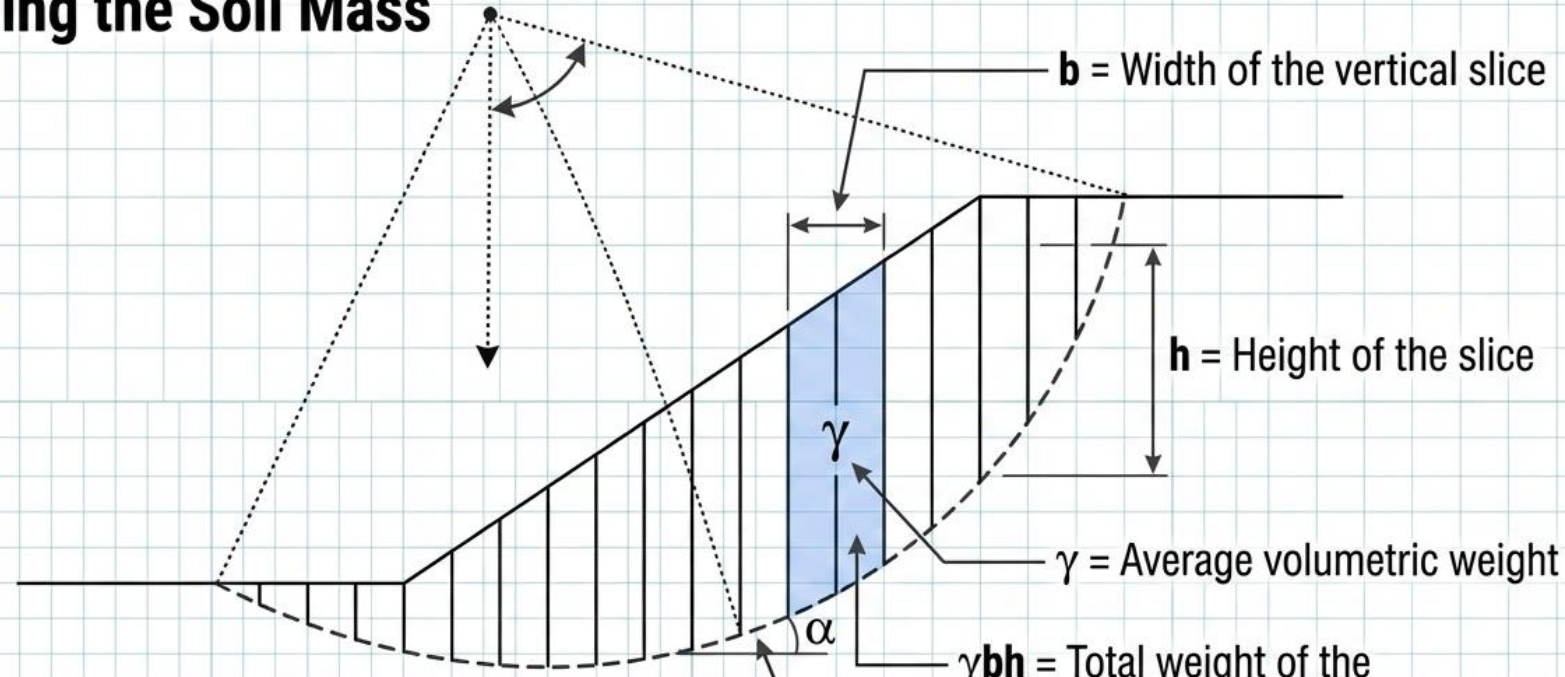

High Effective Normal Stress (*N'*). Particles tightly interlocked. Maximum frictional shear strength.

$$N' = N - u$$

The tipping point isn't more weight; it's water. Acting as an internal hydraulic jack, water mathematically robs the soil of its frictional shear strength.

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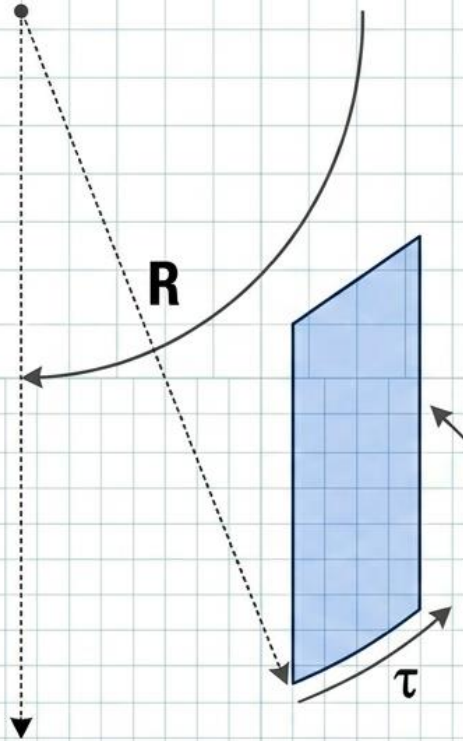
# The Geometric Model: Subdividing the Soil Mass




Bishop's simplified method relies on evaluating the forces acting on these individual vertical subdivisions over an assumed circular slip surface.

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# Establishing Moment Equilibrium Per Slice




$$\sum(\text{Moments of Weight}) = \sum(\text{Moments of Shearing Forces})$$

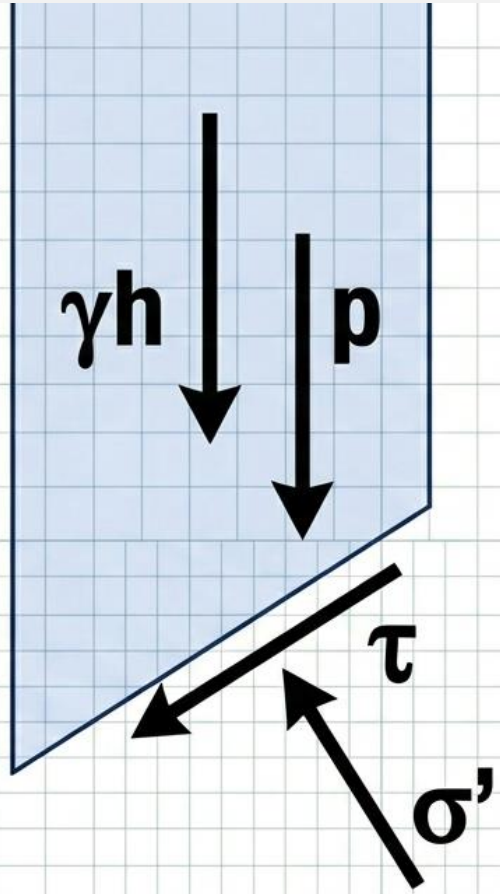
$$\sum \gamma h b R \sin\alpha = \sum \frac{\tau b R}{\cos\alpha}$$

- $\gamma h b$: Weight of slice
- $R \sin\alpha$: Horizontal distance to center (Moment Arm)
- $\tau$: Actual shear stress at the lower boundary

Equilibrium is found by equating moments with respect to the center of the slip circle.

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# The Colving for Normal Effecing Suver ing Normal Effective Stress (σ')




Bishop assumes that forces transmitted between adjacent slices are strictly horizontal.

Vertical equilibrium of a slice yields the expression for normal effective stress:

$$\sigma'\left(1+\frac{\tan\alpha\tan\phi}{F}\right)=\gamma h-p-\frac{c}{F}\tan\alpha$$

Note the presence of the safety factor (F) on the right side of the equation.

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# Deriving the Safety Factor (F) as a Summation

Assuming uniform slice width and applying Coulomb's relation, the stability factor F becomes:

$$F = \frac{\sum[(c + \sigma' \tan \phi) / \cos \alpha]}{\sum \gamma h \sin \alpha}$$

**Normal Effective Stress (σ′) Derivation:**

$$\sigma' \left(1 + \tan \alpha \tan \frac{\phi}{F}\right) = \frac{\gamma h - p - c}{F \tan \alpha}$$

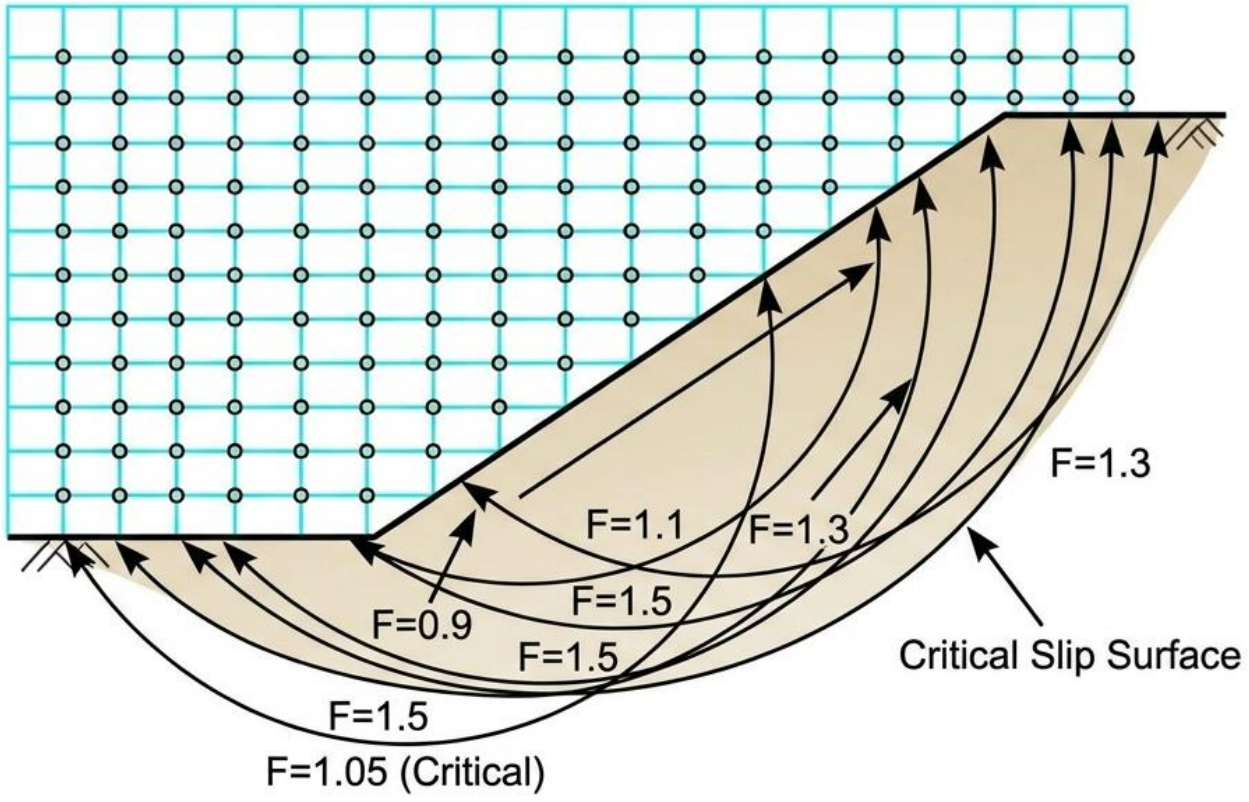
**c** : Local cohesion
**ϕ** : Angle of internal friction
**σ′** : Normal effective stress
**γ** : Average volumetric weight
**h** : Height of the slice
**α** : Angle of the slip surface at the base of the slice
**p** : Forces transmitted between adjacent slices.

This formula assumes the factor **F** is constant across all vertical slices.

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# Slip Surface Analysis: Multiple Slip Circles & Safety Factors

Analysis involves calculating the Safety Factor (F) for numerous potential circular slip surfaces. The critical surface is the one with the lowest F.




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D

# Breakwaters

# Breakwaters- Poseidon's nightmare !



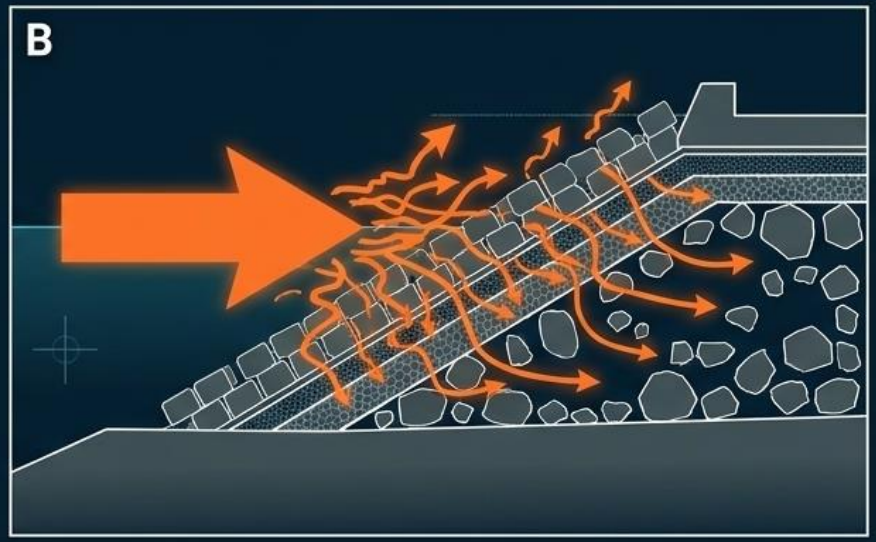
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# THE ENERGY DISSIPATION ENGINE

A rubble mound structure survives not just by deflecting waves, but by swallowing them. Porosity (ranging from 30% to 55%) physically 'chokes' wave energy, fundamentally altering the run-up height ($R_{u2\%}$) and overtopping discharge.




**A. Impermeable Slope**: High Run-up / Surging Waves. $\gamma_f$ = 1.0.




**B. Permeable Armour & Core**: Energy Dissipation through Voids. $\gamma_f$ = 0.40 (Two layers of rock on permeable core).

**DATA INSIGHT**: Permeable cores cap maximum relative run-up ($R_{u2\%}$ / $H_{m0}$) at 2.0, whereas impermeable cores allow run-up to reach nearly 3.0 at large breaker parameters ($\xi_{m-1,0}$).

B

# Why do we need Breakwaters ? Energy Dissipation !









## The Threat

Unabated wave action and tidal currents constantly threaten valuable coastlines.

## The Function

Breakwaters exist to attenuate kinetic wave energy, providing safe harborages, minimizing port operation downtimes, protecting beaches from erosion, and shielding habitats.

## The Stakes

When the shield fails, the port cannot function. Secondary economic damages (loss of production) rapidly outpace the physical repair costs.

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Image Source: https://marineconstructionsource.com/how-do-breakwaters-work/

# The Arsenal: Typologies of Defense



## Rubble Mound

High wave energy dissipation; preferred for reduced reflection.



## Monolithic (Caisson)

Used in deep water where rubble material quantities become uneconomical.

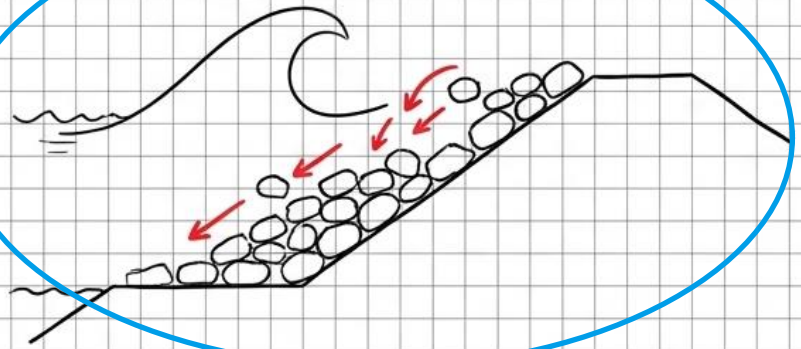


## Floating (Pontoon)

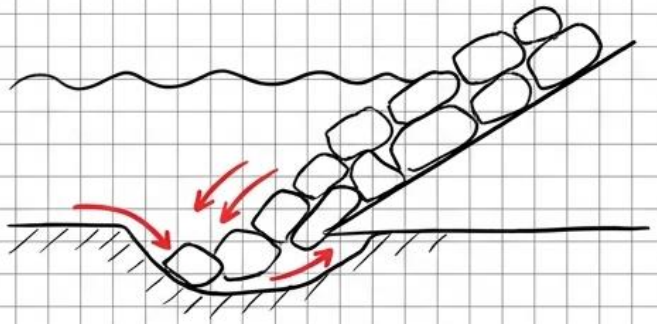
Low-cost, environmentally-friendly alternative for specific wave climates.

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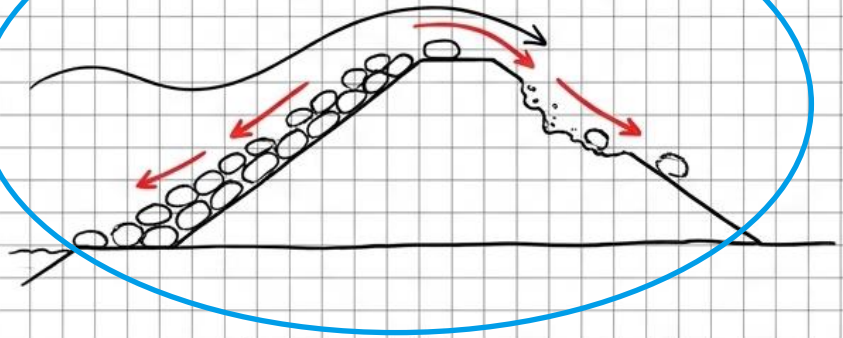
## 1. Armor Layer Instability (Displacement)



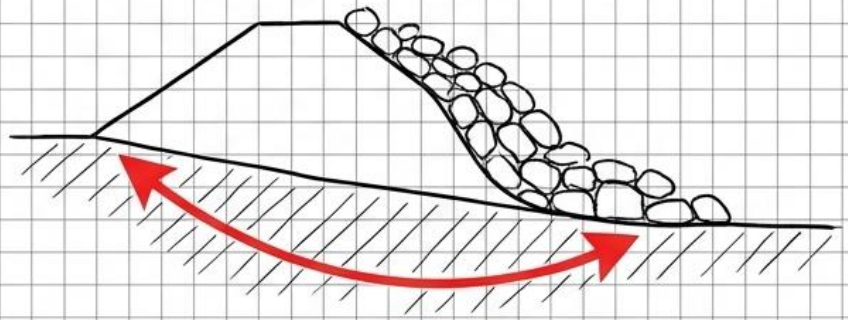
## 2. Toe Scour & Undermining



## 3. Overtopping & Leeward Erosion



## 4. Geotechnical Instability (Slip Circle)



Every element of the cross-section is trying to fail simultaneously. Survival dictates precise mathematical optimization of the armor and the crest.

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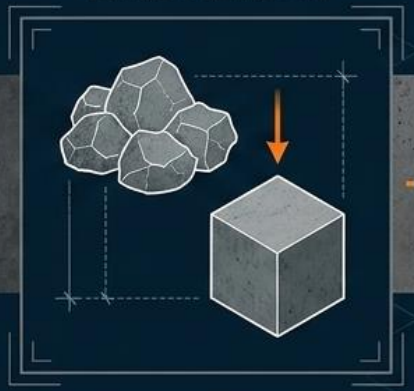
# D.1

# Armour Stability


***Armoring*** *dissipates energy based on:*
- *Local roughness*
- *Tortuosity (seepage paths)*
Caption from video: https://www.youtube.com/watch?v=37qIgRvj-Z4

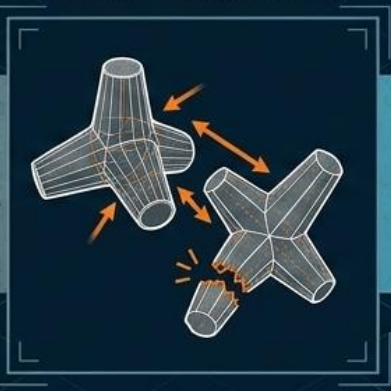
# THE ARMOUR TAXONOMY: FROM MASS TO GEOMETRY

## PHASE 1: PURE MASS
(GRAVITY RELIANCE)



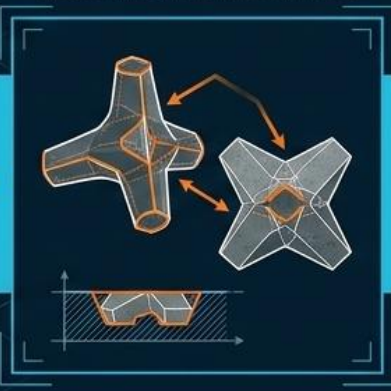
1. Natural Rock: Random placement, heavily reliant on local quarry yields and sheer weight.
2. Cubes / Blocks: The first artificial solution. Massive concrete volume. Smooth surfaces limit interlocking but provide easy placement.

## PHASE 2: COMPLEX FRICTION
(INTERLOCKING RELIANCE)



1. Tetrapods & Dolos: Multi-axis limbs. Designed to knit together in a 3D matrix. High porosity, but vulnerable to limb breakage.

## PHASE 3: COMPUTATIONAL GEOMETRY
(SINGLE-LAYER EFFICIENCY)



1. Accropode, Xbloc, Cubipods: Highly engineered, heavily patented shapes. Designed specifically to maximize frictional interlock and porosity while minimizing the total volume of concrete required.

TU Delft
NotebookLM

# The Master Armour Matrix

| | Placement System | Stability Mechanism | Porosity Factor | Material Efficiency | Logistics |
|---|---|---|---|---|---|
| **Natural Rock** | Random, Multi | Mass | Med | N/A | Local Quarry Limit |
| **Cubes / Blocks** | Random, 2-Layer | Mass | Low | Poor | Easy Cast |
| **Tetrapods / Dolos** | Random/Patterned, 2-Layer | Interlock | High | Moderate | Complex Cast, Fragile |
| **Accropode / Xbloc** | Strict Pattern, 1-Layer | **High Interlock** | High | **Excellent** | Patented, Strict QA |
| **Cubipods** | Flexible, 1 or 2-Layer | Friction/Mass | Med/High | Good | High logistical ease |

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# Hudson Formula: The Basics

Criteria for armor stone stability against wave loads in regular waves (assumes static load).

## Stability Coefficient (Kd)

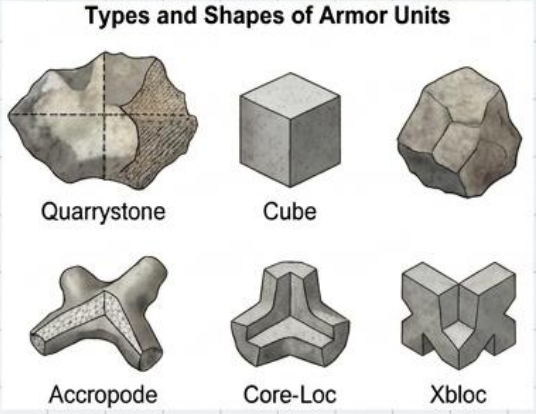
An empirical coefficient accounting for armor unit type, placement, layer thickness, and wave breaking.

$$W = \frac{\rho_r \cdot g \cdot H^3}{K_d \cdot \left(\frac{\rho_r}{\rho_w} - 1\right)^3 \cdot \cot(\alpha)}$$

$$D = \left(\frac{W}{\rho_r \cdot g}\right)^{1/3}$$

## Reference Kd Values (Selected)

| Armor Type (Placement) | Kd Value |
|---|---|
| Quarrystone (Rough, 2 layers) | 3.5 - 4.0 |
| Accropode (Random, 1 layer) | 15 |
| Core-Loc (Random, 1 layer) | 18 |




W: Median weight of an individual armor unit
H: Characteristic wave height (e.g., H10 or Hs)
$\rho_r$: Density of armor unit material
$\rho_w$: Density of water
g: Acceleration due to gravity
$K_d$: Stability coefficient (dimensionless)
α: Slope angle of the structure
D: Diameter of armor unit

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# Van der Meer Formula: The Basics

Criteria for armor stone stability against wave loads (accounting for storm duration, permeability, and wave breaking).

### Damage Level (S)

A non-dimensional parameter representing the eroded area in a cross-section of the armor layer.

$$S = \frac{A}{(D_{n50})^2}$$

### Surf Similarity Parameter ($\xi_{m-1,0}$)

$$\xi_{m-1,0} = \frac{\tan\alpha}{\sqrt{s_{m-1,0}}}$$

### Plunging Waves ($\xi_{m-1,0} < \xi_{mc}$)

$$\frac{H_s}{\Delta D_{n50}} = 6.2 \cdot P^{0.18} \left(\frac{S}{\sqrt{N}}\right)^{0.2} \xi_{m-1,0}^{-0.5}$$

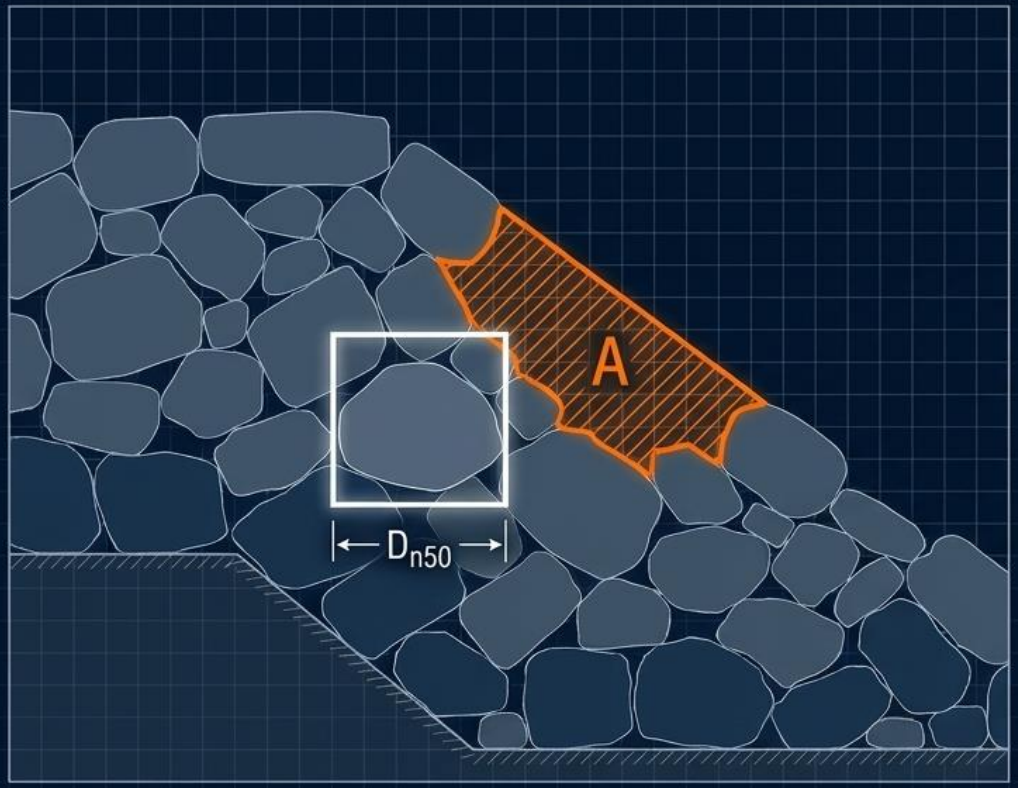
### Surging Waves ($\xi_{m-1,0} > \xi_{mc}$)

$$\frac{H_s}{\Delta D_{n50}} = 1.0 \cdot P^{-0.13} \left(\frac{S}{\sqrt{N}}\right)^{0.2} \xi_{m-1,0}^{P} \cdot \sqrt{\cot\alpha}$$

$H_s$: Significant wave height
$\Delta$: Relative buoyant density of armor unit ($\rho_r/\rho_w - 1$)
$A$: Eroded area in a cross-section of the armor layer
$P$: Notional permeability factor of the structure
$S$: Damage level
$N$: Number of waves in storm duration
$\xi_{m-1,0}$: Surf similarity parameter
$\alpha$: Slope angle of the structure
$s_{m-1,0}$: Wave steepness ($H_{m0}/L_{m-1,0}$) (based on the $T_{-1,0}$ period)

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# Deconstructing the Defense: Mass & Yield




## Parameter Breakdown

**$D_{n50}$ (Nominal Diameter):** The primary unit of resistance. The equivalent cube size of the median armour unit.

**$\Delta$ (Relative Mass Density):** The buoyancy-adjusted weight of the material ($\rho_a/\rho_w$ - 1).

**S (Damage Level):** $S = \frac{A}{D_{n50}^2}$

### Quantifying 'Acceptable' Damage:

S is not an abstract percentage. It is literally the cross-sectional area of erosion (A) normalized by the square of the nominal rock diameter. Turning the S dial defines the maintenance lifecycle.

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# The Core Engineering Dilemma

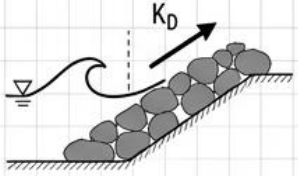
Coastal engineering is not about building an indestructible wall.
It is an economic optimization problem.

## Hudson vs. Van der Meer Stability Equations

### The Hudson Equation (Traditional/Simplified)

$$N_s = \frac{H_s}{\Delta D_{n50}} = (K_D \cdot \cot(\alpha))^{1/3}$$

Based on monochromatic waves, primarily considers wave height and slope angle. Uses a single stability coefficient ($K_D$). Limited coefficenting (P). Limited in accounting for wave period, permeability, and storm duration (number of waves). Often results in conservative, heavier designs.



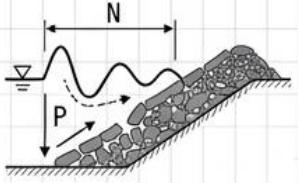

### The Van der Meer Equation (Modern/Comprehensive)

$$N_s = \frac{H_s}{\Delta D_{n50}} = 6.2 \cdot P^{0.18} \left(\frac{S}{\sqrt{N}}\right)^{0.2} \cdot \xi_m^{-0.5}$$

or

$$N_s = 1.0 \cdot P^{-0.13} \left(\frac{S}{\sqrt{N}}\right)^{0.2} \cdot \cot(\alpha)^{0.5} \cdot \xi_m^{P}$$

Accounts for irregular waves, storm duration (N), structural permeability (P), breaker parameter ($\xi_m$), and allowable damage level (S). Provides a more detailed and optimized design, allowing for calculated risk.



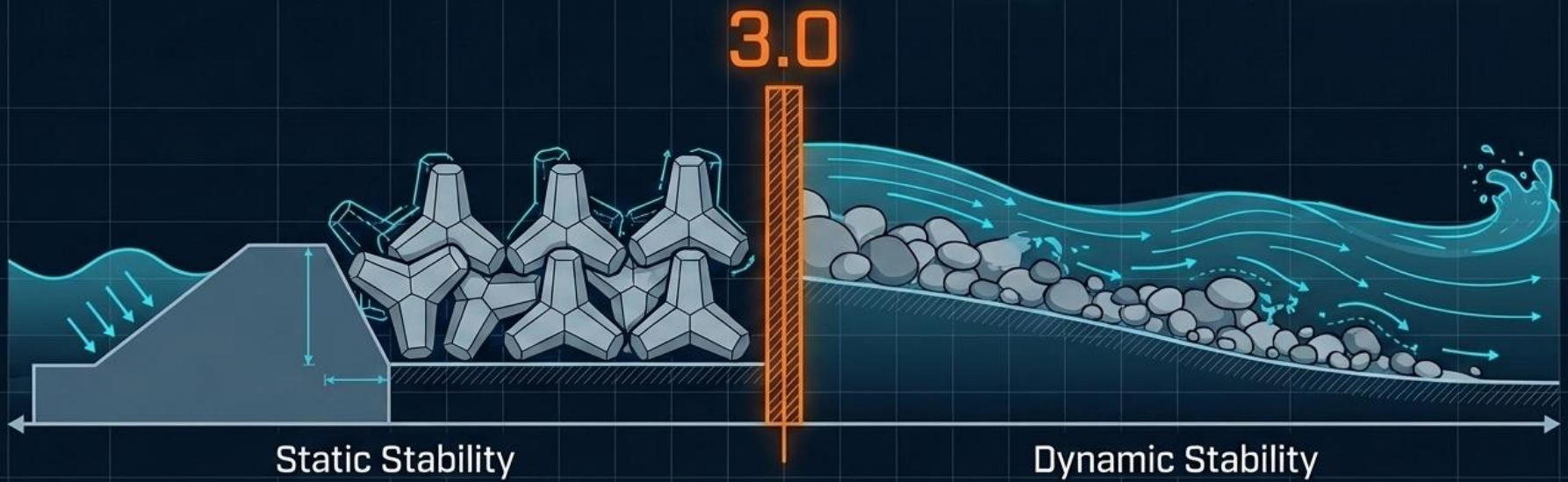

**Takeaway: The Van der Meer formulas exist precisely to quantify and navigate this dilemma.**

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# The Stability Threshold: Static vs. Dynamic

$$\frac{H_{m0}}{\Delta D_{n50}} = 3.0$$

$N = 2.-$ (Van der Meer)




**Body**

A breakwater is designed to be statically stable (allowing for minor initial reshaping). If the design pushes the stability number (Hm0 / ΔDn50) beyond 3.0 for 100-year wave conditions, the structure crosses a hydrodynamic threshold.

**Takeaway**

Beyond 3.0, the structure is no longer classified as a breakwater. It acts as a dynamically stable structure (like a rock or shingle beach) governed by entirely different rules, including longshore transport.

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# D.3

# Overtopping


***Breakwaters*** *are designed to be overtopped depending on what its behind them.*
*$Ru_{2\%}$ is 2 out of a 100 waves.*
*The q depends on how big are the 2% the waves !*
Caption from video: https://www.youtube.com/watch?v=37qIgRvj-Z4

# EurOtop Overtopping: Design Criteria (q & $V_{max}$) & Allowable Values

Formulas for calculating key breakwater overtopping metrics according to the EurOtop manual.

## Mean Discharge (q)

$$\frac{q}{\sqrt{gH_{m0}^3}} = 0.09 \cdot \exp\left(-1.5 \frac{R_c}{H_{m0}\gamma_f\gamma_\beta}\right)$$

The influence factor $\gamma_{BB}$ can be described by Equations 6.11 and 6.12.

| | | |
|---|---|---|
| $\gamma_{BB} = 0.68 - 4.1s_{m-1,0} - 0.05B/H_{m0}$ | for hardly and partly reshaping berm breakwaters | 6.11 |
| $\gamma_{BB} = 0.70 - 8.2s_{m-1,0}$ | for fully reshaping berm breakwaters | 6.12 |

## Max Volume ($V_{max}$)

$$V_{max} = V_{bar} \cdot (\ln(N_{ow}))^{0.75}$$

$$\text{where } V_{bar} = q\frac{T_{m-1,0}}{N_{ow}}$$

Represents the hazard from a single wave. Key for assessing immediate danger to people and structural integrity.

## Influence Factors ($\gamma_b$ and $\gamma_f$):

| Berm Influence Factor ($\gamma_b$) | Roughness Influence Factor ($\gamma_f$) |
|---|---|
| $\gamma_b = 1 - r_B(1 - \gamma_{bb})$ | $\gamma_f = f(\text{roughness element type, placement, slope angle})$ |
| Accounts for the effect of a berm on run-up and overtopping. Reduces wave energy by dissipation and by acting as a buffer. | Represents the effect of slope roughness on wave run-up and overtopping. Increased roughness leads to higher energy dissipation. |

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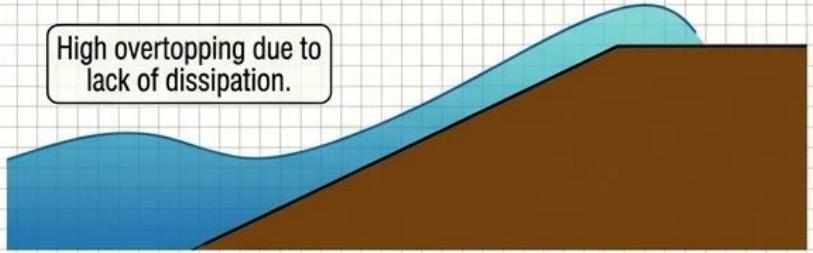
| Return period | Tolerable discharge | Comment | Reference |
|---|---|---|---|
| 1 in 1 year | q<1.0 l/s per m, $V_{max}$ < 600 l/m. | For people and vehicles - People at seawall / dike crest. Clear view of the sea. $H_{m0}$ = 2 m. | EurOtop (2018) Table 3.3 |
| 1 in 1 year | q<1.0 l/s per m $V_{max}$ < 500 l/m. | For pedestrians – Trained staff, well shod and protected, expecting to get wet, overtopping flows at lower levels only, no falling jet, low danger of fall from walkway. | EurOtop (2007): Table 3.2 |
| 1 in 200 year | q<5.0 l/s per m, $V_{max}$ < 2,000-3,000 l/m | For structural design - Grass covered crest and landward slope; maintained and closed grass cover; $H_{m0}$ = 1 – 3 m | EurOtop (2018) Table 3.1 |
| 1 in 200 year | q<10 l/s per m | For damage to defence – No damage to crest and rear face of embankment of clay. | EurOtop (2007): Table 3.5 |

$$V_{\max} = a \cdot [\ln(N_{ow})]^{1/b} \qquad 4.6$$

# EurOtop Overtopping: Influence Factors - Visualized

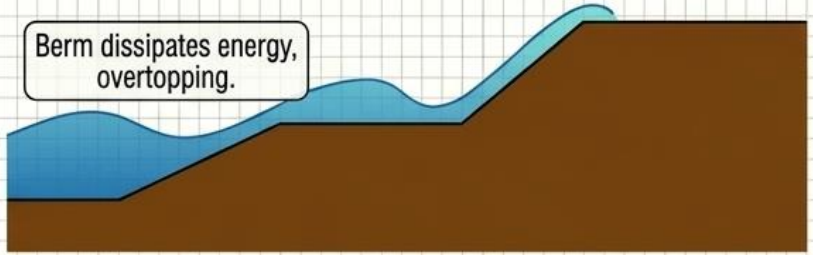
## Berm Influence Factor ($\gamma_b$)

### No Berm ($\gamma_b \approx 1$)




High overtopping due to lack of dissipation.

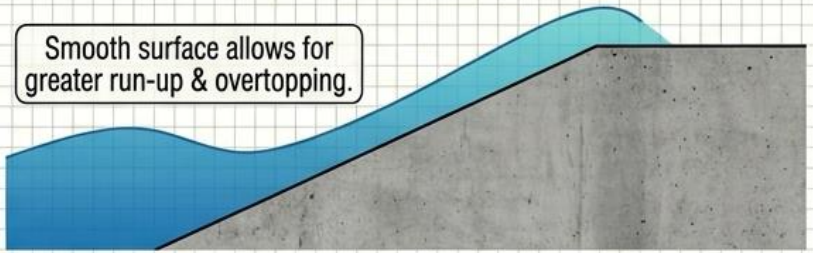
### With Berm ($\gamma_b < 1$)




Berm dissipates energy, reducing run-up & overtopping.

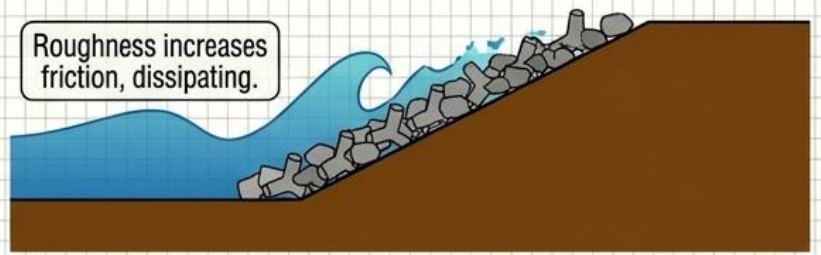
## Roughness Influence Factor ($\gamma_f$)

### Smooth Slope ($\gamma_f \approx 1$)




Smooth surface allows for greater run-up & overtopping.

### Rough Slope ($\gamma_f < 1$)




Roughness increases friction, dissipating wave energy

E

# Bed Protection

# ***This is not Bed Protection !***



https://www.amazon.com/Waterproof-Protector-Absorbent-Breathable-Hypoallergenic/dp/B07H33JHJH?th=1

# This is Bed Protection !




https://www.amazon.com/Waterproof-Protector-Absorbent-Breathable-Hypoallergenic/dp/B07H33JHJH?th=1

## The Core Challenge: Flow vs. Form

Water relentlessly seeks to move earth. Our engineering objective **is not to fight the flow, but to design a geometric matrix that absorbs shear stress while remaining structurally static.**

The solution scales across three dimensions:

1. **Particle** (Initiation of motion)
2. **Slope** (Gravitational effects)
3. **System** (Multi-layered filtration)




**Shear Velocity ($u^*$):** This 'friction velocity' is a theoretical value derived from the bed shear stress ($\tau_b$) and fluid density ($\rho$), calculated as $u_* = \sqrt{(\tau_b/\rho)}$. It measures the intensity of the turbulent flow's drag on the bed.

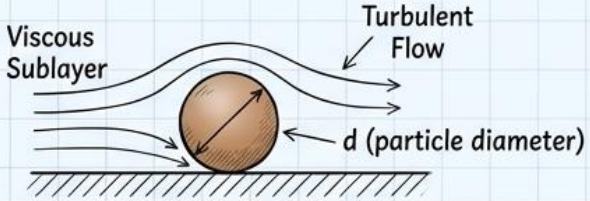
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# Particle Reynolds Number (Re*): Explanation and Formula

*A dimensionless quantity characterizing flow near bed particles.*

## Definition & Significance

Re* relates inertial forces to viscous forces at the particle scale. It characterizes the flow regime (laminar, transitional, turbulent) around individual sediment particles.




## The Formula

$$\mathbf{Re^*} = \frac{u^* \cdot d}{\nu}$$

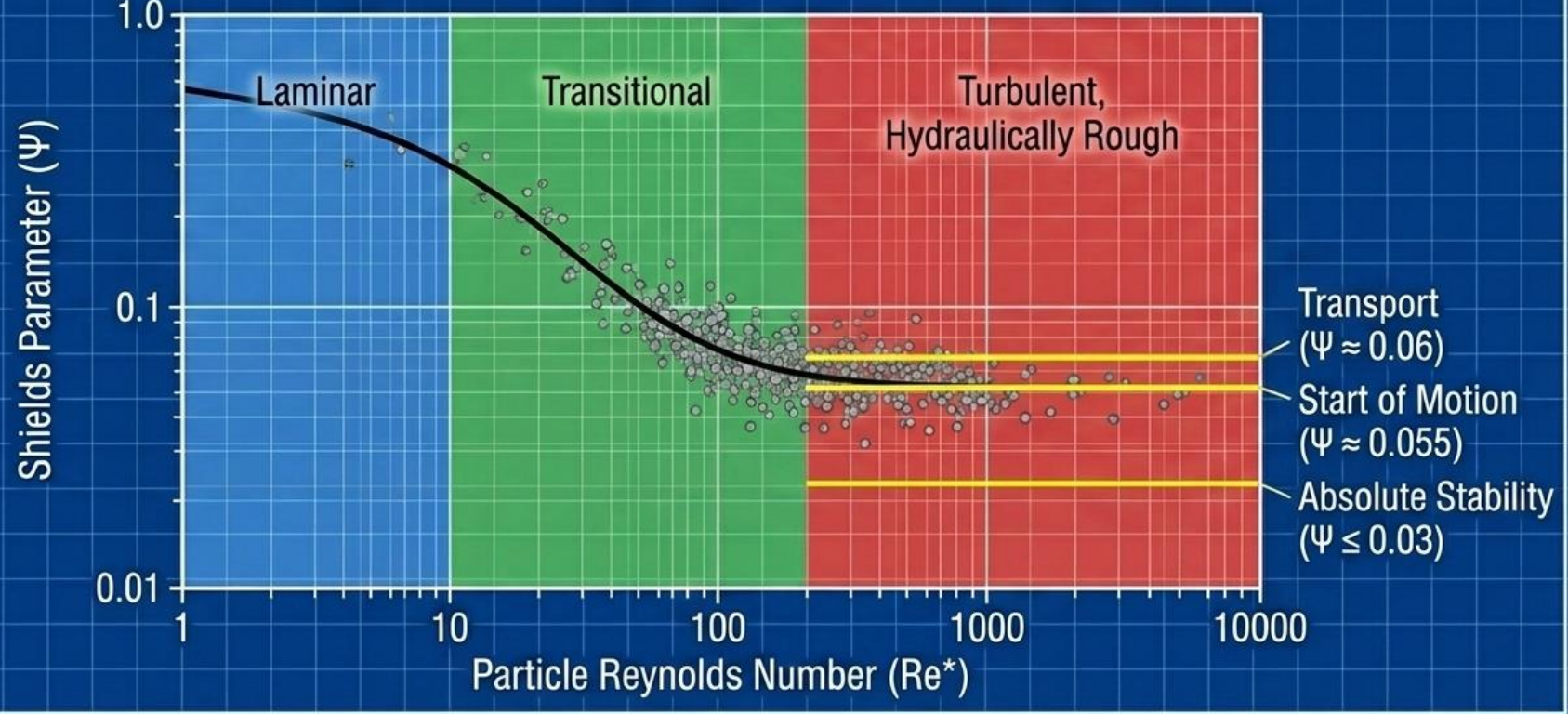
- $u^*$ = Friction Velocity (shear velocity)
- $d$ = Particle Diameter
- $\nu$ = Kinematic Viscosity of the Fluid




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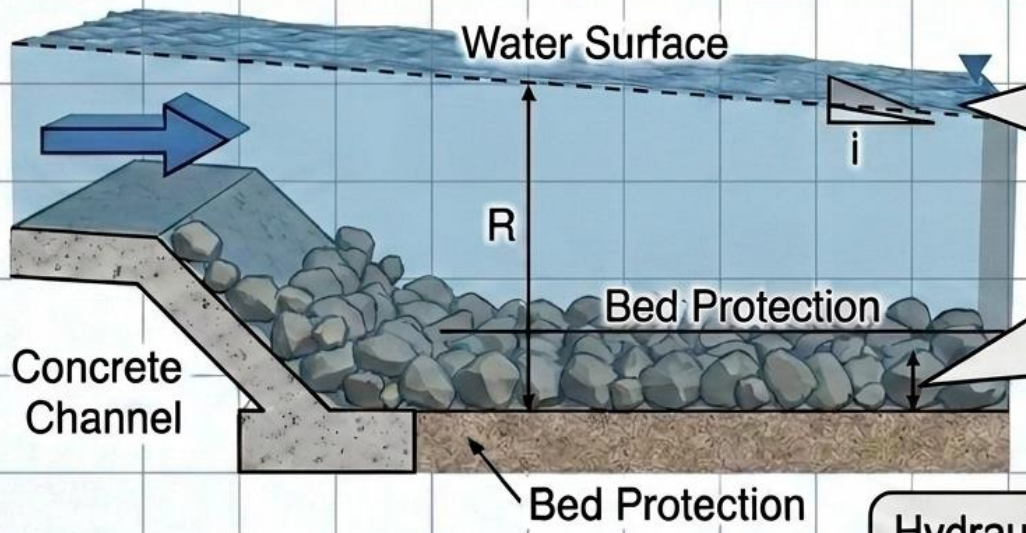
# Defining Critical Limits: The Safety Threshold

## Critical Threshold Lines on the Shields Diagram




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# Visualizing the Variables: Bed Protection Example




**Shields Parameter Formula ($\Psi$)**

$$\Psi = \frac{R \cdot i}{\Delta \cdot g \cdot D}$$

**D (Particle Diameter)**



$\Delta = (\rho_s - \rho_w)/\rho_w$
(Relative Density)

R = Hydraulic Radius
i = Energy Slope
$\Delta$ = Relative submerged density
D = Characteristic particle diameter

**Hydraulic Radius Formulas (R = A/P)**

Rectangular Channel: $R = \frac{b*h}{b+2h}$

Trapezoidal Channel: $R = \frac{(b+zh)h}{b+2h\sqrt{1+z^2}}$

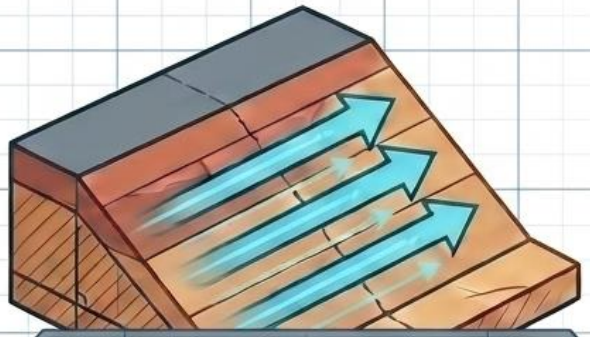
Circular Pipe (Full): $R = \frac{D}{4}$

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# Effective Shields Parameter

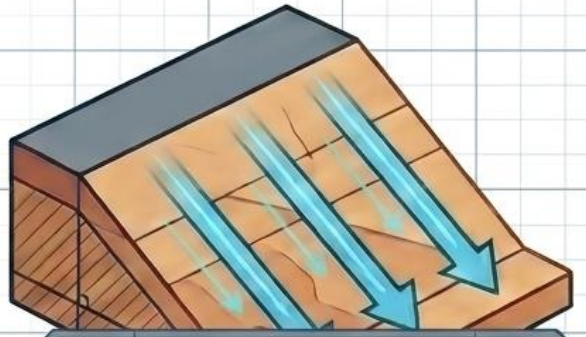
Bank angles ($\alpha$) actively assist fluid shear stress. To maintain our safety threshold, the critical Shields parameter must be reduced by a factor of $k_\alpha$, dependent strictly on flow trajectory relative to the slope angle ($\phi$).

## Flow Parallel to Depth Lines



$$k_\alpha = \cos(\alpha) * \left(1 - \frac{\tan(\alpha)}{\tan(\phi)}\right)$$

## Flow Perpendicular to Depth Lines



$$k_\alpha = \cos(\alpha) * \sqrt{1 - \frac{\tan^2(\alpha)}{\tan^2(\phi)}}$$

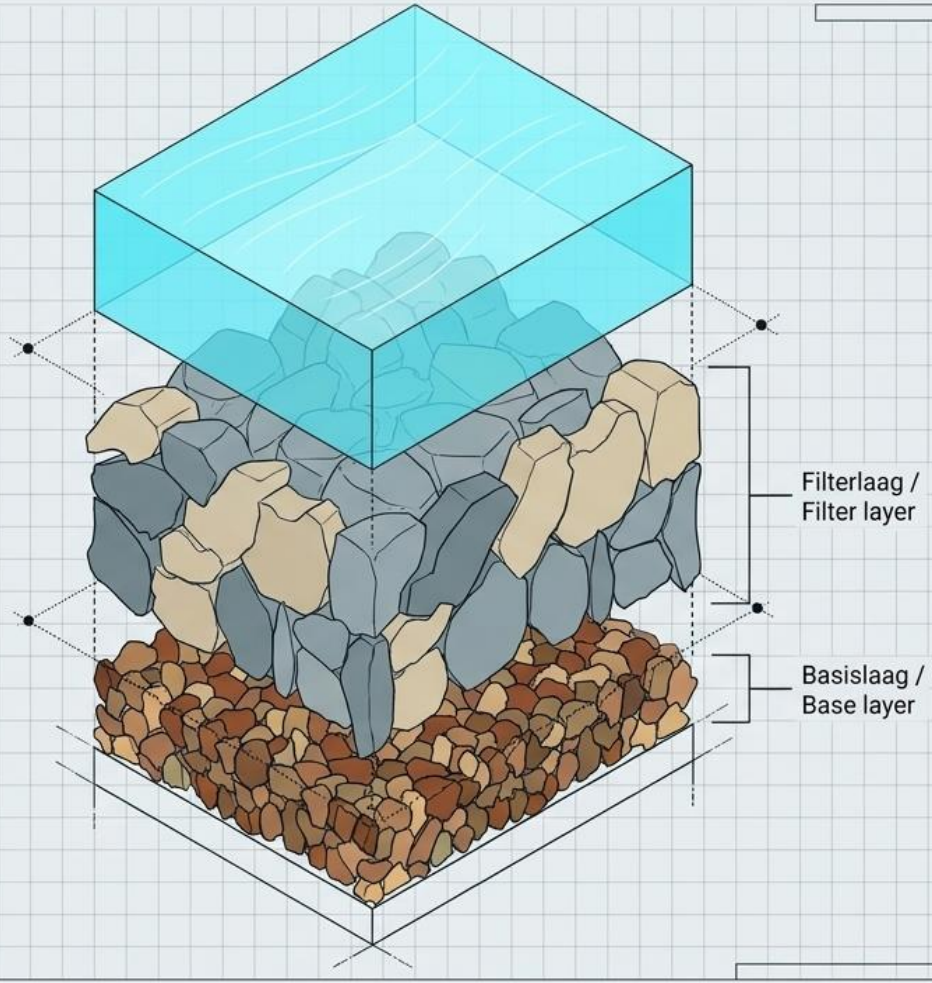
The effective critical Shields parameter on a slope is calculated as:

$$\Psi_{cr,\alpha} = k_\alpha \cdot \Psi_{cr,0}$$

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## The System Scale: Granular Filters

To prevent the washing out of fine subsoils while withstanding high external shear stress, single-layer protection is insufficient. We must engineer geometrically calibrated transition layers.




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# Granular Sizing: The Graded Material Rule

Because natural rock is un-uniform, the diameter D in our equations must represent the median equivalent sphere.




$$D = D_{n,50} = \left( \frac{m_{50}}{\rho_s} \right)^{1/3}$$

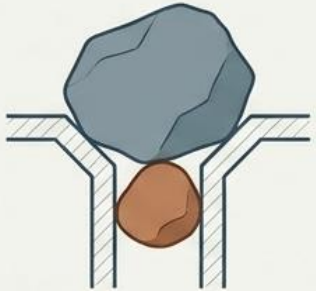
$m_{50}$ represents the mass exceeded by exactly 50% of the graded grains.

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# The Three Pillars of Filter Matrix Design

*(Subscripts: f = filter, b = base, n = same layer)*

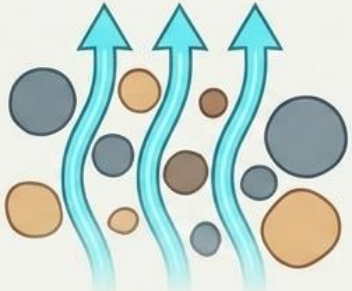
## 1. Retention (Geometrically Closed)



Goal: Prevent base material from migrating through the filter.

$$\frac{D(15, f)}{D(85, b)} < 5$$

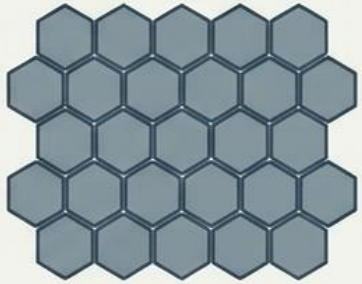
## 2. Permeability (Waterdoorlatend)



Goal: Ensure water pressure can escape without building up behind the filter.

$$\frac{D(15, f)}{D(15, b)} > 5$$

## 3. Internal Stability



Goal: Ensure the layer itself does not self-segregate or collapse.

$$\frac{D(60, n)}{D(10, n)} < 10$$

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F

# Conclusions

# Dikes and Breakwaters

- Dikes and Breakwaters have many ways of failing
- Failure is nicelly expressesed in load and resistance.
- Design of structures is largelly about energy dissipation.
- As civil engineers, we choose the geometry and the materials to ensure resistance to a given load.